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MEASURING BARRIERS TO FUSION

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ABSTRACT

The experimental extraction of detailed barrier distributions has brought a significant advance in the study of the fusion of heavy nuclei, and indeed in the entire heavy-ion reaction process. A quantitative understanding of the entrance-channel effects induced by target and projectile structure has emerged, based on recent high-precision measurements of fusion excitation functions. These distributions show clearly whether the experimental data are good enough to give the information required. They are also the functions best suited to the theoretical interpretation of the reaction dynamics—often presenting an unambiguous "fingerprint" of the target and projectile structure. We are now at the stage where we can start to exploit the insights gained in order to understand properties of the compound nucleus created: its spin distribution and evaporation residues, perhaps its possible shapes, and, in the case of heavy systems, its subsequent fission.

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1. INTRODUCTION

A fundamental aspect of quantum mechanics is the tunnelling of a particle into or through a classically forbidden region. The first experimental proof of this phenomenon came from nuclear alpha decay (1). The analagous process of nuclear fusion was described similarly, in terms of a simple Coulomb barrier with the radial motion being the only degree of freedom. Low-energy fusion probabilities for light nuclei, particularly in nucleosynthesis, have been successfully described (2) in this way. However, in general, the participants have internal degrees of freedom, and couplings to these can dramatically modify the tunnelling probability. The phenomenon of coupling-assisted tunnelling is a general one, appearing in such diverse processes as catalysis of chemical

reactions, behavior of solid-state junctions, and atomic and nuclear collisions (3).

In nuclear collisions, the most dramatic consequence of coupling-assisted tunnelling occurs in low-energy heavy-ion fusion (4, 5), where cross sections are enhanced by orders of magnitude as a result of coupling to collective modes of the interacting nuclei. Under certain approximations, the single fusion barrier can be thought of as splitting into a distribution of barriers in the presence of couplings. The barrier energies and probabilities are characteristic of the couplings and can be related to the configurations of the colliding nuclei at contact. It was believed, however, that fusion cross sections, while significantly affected by these couplings, varied smoothly with beam energy and could not be used to investigate directly the specific nature of these effects (6).

Rowley et al (7) proposed in 1991 that the distribution of barriers could be extracted directly from precise fusion cross section measurements through a simple and elegant mathematical transformation. Subsequent measurements of fusion barrier distributions have clearly shown characteristic features reflecting the couplings specific to the internal structure of the colliding nuclei, demonstrating a new richness in this phenomenon. Many other channels are influenced by the same sets of barriers as the fusion; indeed, they govern the whole dynamics of the reaction in the barrier region, although their consequences are generally most apparent in the fusion cross section, since this is essentially an incoherent sum over the individual barrier contributions. The understanding of barrier distributions acquired in the past few years can now be exploited in studying a wide range of phenomena, for example, fission probabilities and anisotropies and the population of nuclei at large deformation or far from the line of beta stability.

Most of the concepts in this review apply to collisions at energies close to the Coulomb barrier, where the nuclei come together relatively slowly and do not overlap significantly before the barrier is passed. Earlier reviews on the subject of heavy-ion fusion near the Coulomb barrier are available (4, 5, 8, 9).

The concept of a barrier distribution applies even when the couplings are weak; however, the width of the distribution, ΔB, is proportional to $Z_1Z_2\beta R$, where βR is the relevant deformation length and Z_1Z_2 is the charge product. The barrier distributions determined for $^{16}O + {}^{58,62}Ni$ (10) show that barrier structure is very difficult to measure for systems with Z_1Z_2 less than about 250. Thus, in the context of our problem, we define "light" systems as those having Z_1Z_2 less than 250 and do not discuss them further in this review. At the other end of the Z_1Z_2 scale, we regard as "heavy" those systems having Z_1Z_2 greater than about 1600. For these systems, frictional effects start to become important, and as discussed in Section 12, the entire fusion process then becomes more complex. This review concentrates on the concept and

experimental determination of fusion barrier distributions in systems between the light and heavy ones defined above.

The rest of this section is devoted to the simplest model of fusion, namely, the single-barrier model (or one-dimensional barrier penetration model). In Section 2 we discuss the reasons behind the failure of this simple model. Sections 3 and 4 then develop the idea of the fusion barrier distribution and the consequences of coupling to other degrees of freedom. Experimental methods and the need for precise measurements are addressed in Section 5, followed in Sections 6 through 8 by discussions of experimental barrier distributions for reactions that highlight specific couplings. Section 9 considers couplings to high-lying states, and Section 10 focuses on attempts to extract barrier distributions from complementary scattering data. The implications of our better understanding of fusion through barrier distributions are discussed in section 11, with emphasis on applications to the fission process. Section 12 examines reactions in which passage over the fusion barrier does not necessarily imply formation of an equilibrated compound nucleus. The current status and future prospects for fusion of radioactive nuclei are discussed briefly in section 13, followed by a summary and outlook in the last section.

1.1 *Single-Barrier Penetration Model*

The basic features of nuclear reactions are commonly described in terms of an interaction that is a function of the distance between the centers-of-mass of the target and projectile and consists of a repulsive Coulomb term and a short-ranged attractive nuclear component. The total potential possesses a maximum at a distance where the repulsive and attractive forces balance each other. This is referred to as the Coulomb barrier, and the energy of relative motion must be sufficient to overcome this barrier in order for the nuclei to be captured and fuse. The larger the charge product $Z_1 Z_2$ of the two nuclei, the greater is the repulsive Coulomb force, and thus a larger overlap of the nuclear densities is attained before the barrier is reached, as illustrated in Figure 1.

The presence of a pocket in the nucleus-nucleus potential allows a simple conceptual criterion for fusion: If the system enters the pocket, it is captured and fuses. Details of the dynamical processes forming the fully equilibrated compound nucleus are unnecessary, although the general picture of the processes that cause capture inside the barrier is fairly clear. Once significant density overlap occurs, nucleon-nucleon interactions become prolific, resulting in a substantial loss of kinetic energy and angular momentum from the relative motion, so that the nuclei can no longer escape from the potential well and all the flux passing the barrier leads to fusion. Within this single-barrier passing model, alternatively known as the one-dimensional barrier penetration model (because the separation is the only degree of freedom), the fusion cross section

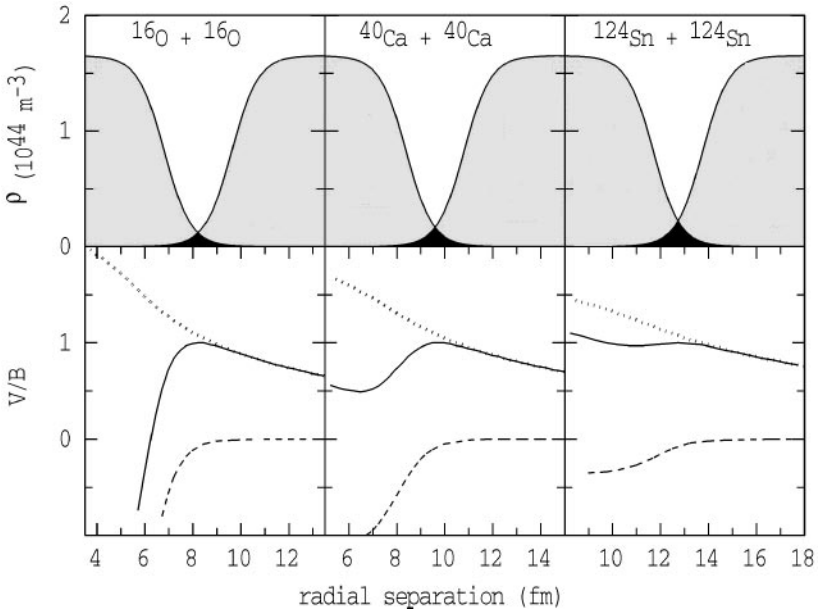


Figure 1 *Lower panels*: The Coulomb (*dotted lines*), the nuclear (*dashed lines*), and the total interaction potentials (*solid lines*), as a ratio to the fusion barrier energy B, for systems with charge products $Z_1Z_2 = 64$, 400, and 2500. The total potential is shown for orbital angular momentum $l = 0$ only. *Upper panels*: The corresponding nucleon density distributions at the barrier.

σ at an energy E is given by a summation over partial waves:

$$\sigma = \frac{\pi\hbar^2}{2\mu E}\sum_{l=0}^{\infty}(2l+1)T_l(E), \qquad 1.$$

where μ is the reduced mass of the system and T_l is the barrier transmission probability.

The above hypothesis works well for kinetic energies that are not too far above the fusion barrier, but at higher energies, barriers for higher angular momenta become relevant and the potential pocket may eventually disappear [or for light systems the excitation energy may even lie below the compound nucleus yrast line (i.e. no states exist for the corresponding spin), preventing fusion (11)]. The frictional and nuclear breakup processes that are invoked to describe fusion and incomplete fusion in this regime (12) are not discussed in this review, which concentrates on fusion at energies close to the Coulomb barrier.

2. BEYOND THE SINGLE-BARRIER MODEL

The single-barrier penetration model successfully described fusion cross sections for light nuclei (13). However, cross sections for heavier systems were found to be significantly larger than expected at energies below the calculated barrier, where enhancements of several orders of magnitude were observed (4, 5). Balantekin et al (14) elegantly showed the limitations of the simple model, extracting an effective potential by inverting the experimental data. For the heavier systems this procedure yielded unphysical internuclear potentials, while those for lighter ions matched well with phenomenological models. This demonstrated conclusively that the discrepancies were not due to the use of the wrong form of potential; rather the assumption of inert, spherical nuclei interacting through an effective one-dimensional potential was no longer adequate. A more microscopic description was required, incorporating the physical processes that influence fusion. In the problem of alpha decay, too, it has recently been found essential to take account of the internal structure of the daughter in a detailed description of the decay process (15).

2.1 *Experimental Evidence for the Role of Internal Structure*

The seminal work of Stokstad et al (16) provided experimental evidence that the structure of the interacting nuclei has a dramatic effect on heavy-ion fusion cross sections. Measurements for $^{16}\mathrm{O} + {}^{148,150,152,154}\mathrm{Sm}$ showed a dramatic increase of sub-barrier fusion cross sections with increasing neutron number. This increase was also observed in the later measurements for $^{40}\mathrm{Ar} + {}^{144,148,154}\mathrm{Sm}$ (17). This dependence is illustrated in Figure 2*a*, which shows the recent, more precise data for $^{16}\mathrm{O} + {}^{144,148,154}\mathrm{Sm}$ (18). Such a strong dependence on isotope masses could not be explained by single-barrier calculations, where the potential depends on the nuclear separation alone. However, the increased cross sections for deformed nuclei such as $^{154}\mathrm{Sm}$ are easy to understand geometrically. Compared with the spherical problem, the Coulomb barrier is lower when the projectile approaches the "pole" of the deformed Sm target and is higher when it approaches the "equator." Taking into account all possible orientations produces a distribution of barrier heights, some lower and some higher than the single barrier. The lower barriers give rise to large cross sections at energies below the original barrier. From this simple semiclassical picture it is clear that the interaction between $^{16}\mathrm{O}$ and $^{154}\mathrm{Sm}$ must include at least one other degree of freedom, namely the orientation of the deformed target.

In a key experiment by Beckerman et al (19), the fusion cross sections for the $^{58}\mathrm{Ni} + {}^{58}\mathrm{Ni}$, $^{58}\mathrm{Ni} + {}^{64}\mathrm{Ni}$, and $^{64}\mathrm{Ni} + {}^{64}\mathrm{Ni}$ reactions, while disagreeing with predictions of single-barrier calculations, showed another interesting feature.

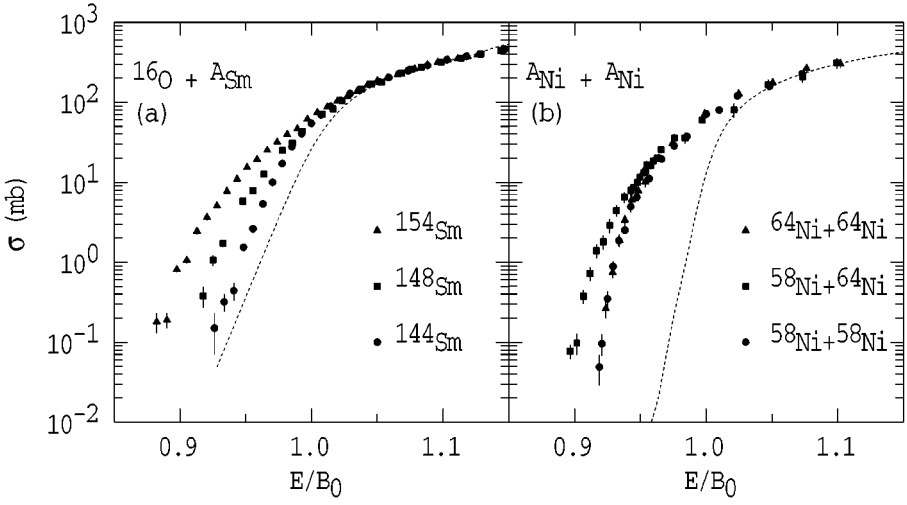


Figure 2 (*a*) Fusion excitation functions for ^{16}O on Sm isotopes (18) show a marked increase of cross section with increasing mass number and deformation. (*b*) The difference in the energy dependence of the cross sections (19) for the ^{58}Ni + ^{58}Ni, ^{58}Ni + ^{64}Ni, and ^{64}Ni + ^{64}Ni reactions indicates that transfer of nucleons affects the fusion process. The *dashed lines* show calculated cross sections assuming a single barrier at energy B_0. The values of E/B_0 in (*b*) have been calculated scaling the B_0 values of Reference 19 by 1.037 to match more recent data.

The excitation function for ^{58}Ni + ^{64}Ni at energies below the Coulomb barrier was found to decrease more gradually than for the other two systems, a behavior that again could not be explained by simple scaling of the isotope sizes. Later data (20) show some quantitative disagreement but confirm this feature, which is illustrated in Figure 2*b*. The fact that among the three systems, ^{58}Ni + ^{64}Ni is the only one with a positive *Q*-value transfer reaction, led to the proposal (21) that specific reaction channels can affect the fusion process.

By the early 1980s there was much compelling evidence that fusion near the barrier is influenced by intrinsic degrees of freedom of the interacting nuclei, primarily deformation or low-lying collective excitations but also the exchange of nucleons between the two nuclei.

2.2 *Calculations Including Nuclear Structure Effects*

Because observed enhancements were not restricted to reactions with deformed nuclei, degrees of freedom other than static deformations were also invoked. These were vibrational modes (22, 23), transfer processes (19, 21), and neck formation (24). For an infinite set of degenerate phonon states, surface vibrations were shown (25) to give rise to a Gaussian distribution of barrier heights. Thus

the experimental excitation functions were calculated by (*a*) averaging over all orientations in the deformed case or (*b*) averaging over all surface separations in the vibrational case. The cross sections were calculated at energy E by introducing a distribution of barrier heights $\mathrm{D}(B)$, with $\sigma(E) = \int D(B)\sigma(B, E)\, dB$. In earlier parameterizations, $\mathrm{D}(B)$ had been sometimes assumed to be rectangular (26), whereas later work found that truncated Gaussian distributions gave better fits for some systems (23). Keller et al (27) made the first attempt to extract the distribution of barrier heights directly from fusion data, using an unfolding procedure. The distributions were reasonably well defined at low energies, but around the peak of the distribution, near the average barrier, the uncertainties were larger than the measured values.

A quantitative justification of the averaging procedure outlined above, along with its limitations, was provided by Jacobs & Smilansky (28) using the framework of the coupled-channels formalism. In a similar approach, Dasso et al (29, 30) provided a formal basis for incorporating all the different degrees of freedom. Through this approach, detailed in Section 4.2, they showed that increased cross sections at low energies are a natural consequence of any couplings to other reaction channels.

3. THEORETICAL INTRODUCTION TO FUSION BARRIER DISTRIBUTIONS

3.1 *The Theoretical Concept and its Experimental Realization*

In the case of a rotational nucleus, where the classical picture of a deformed object oriented in different directions in space is appropriate, it is easy to appreciate the existence of a distribution of fusion barriers. Several important questions are, however, immediately raised:

1. Can meaningful barrier distributions be determined experimentally?
2. If a distribution exists, to what do the heights and weights (probabilities) of the barriers correspond?
3. How does the quantum mechanical description of the deformed rotor, through couplings to members of the rotational band, correspond to the simple classical picture?
4. What happens for other systems where vibrational couplings and/or transfer channels are important and our classical intuition is less clear?

This section addresses some of these questions. Others will be further developed in Section 4. Many detailed theoretical aspects of the problem, as well

as possible approximation schemes, are reviewed elsewhere (31); this review concentrates on the points most directly important to interpreting and understanding experimental data in terms of barrier distributions.

A remarkably simple answer to the first question has emerged (7). An "experimental barrier distribution" can be obtained from the data through the relation

$$D_{\rm exp}(E) = \frac{d^2(E\sigma)}{dE^2}, \qquad 2.$$

where E is the incident energy and σ is the measured fusion cross section. This necessitates taking a second derivative of the experimental data, which imposes very severe constraints on their precision. The present review emphasizes those data that are able to meet these precision requirements.

Figure 3 shows the above quantity for six rather different combinations of target and projectile (18, 32–34), most of which are discussed in more detail below. The figure demonstrates that, unlike the fusion cross section itself, $D_{\rm exp}$ clearly shows significant differences from system to system and in many cases displays detailed structures that can serve as a fingerprint of the nuclear structure of the target and projectile or of the reaction dynamics. The reaction ^{40}Ca + ^{40}Ca (32) can be considered the simplest case, since it features a spherical double closed-shell nucleus as both target and projectile. We therefore expect both inelastic and transfer couplings to have little effect on the fusion barrier distribution, which should appear as a single barrier. The narrow peak in $D_{\rm exp}$ reflects this perfectly. However, this system requires a stronger nuclear potential than expected (9, 32), which has been attributed to polarization; this is discussed in terms of a coupled-channels approach in Section 9.

It is necessary to understand the form of this single peak observed for ^{40}Ca + ^{40}Ca in order to see how Equation 2 comes about and how $D_{\rm exp}$ should be interpreted in other cases. Consider first a single parabolic potential barrier of height B. The barrier transmission for the partial wave l at an energy E may be written as follows:

$$T_l(E) = \frac{1}{1 + \exp[(2\pi/\hbar\omega)(B - E + \hbar^2 l(l+1)/2\mu R^2)]}. \qquad 3.$$

This is the usual Hill-Wheeler (35) expression with the barrier height modified to include a centrifugal term. In the above expression, μ is the reduced mass of the system, R is the barrier radius, and ω is the oscillator frequency of the inverted barrier. The corresponding fusion cross section is given by inserting this expression for T_l into Equation 1. Replacing the sum in that equation by an integral, it is relatively simple to derive the following expression (36):

$$E\sigma = \frac{\hbar\omega R^2}{2}\ln(1 + \exp[2\pi(E - B)/\hbar\omega]) \equiv \frac{\hbar\omega R^2}{2}\ln(1 + e^x), \qquad 4.$$

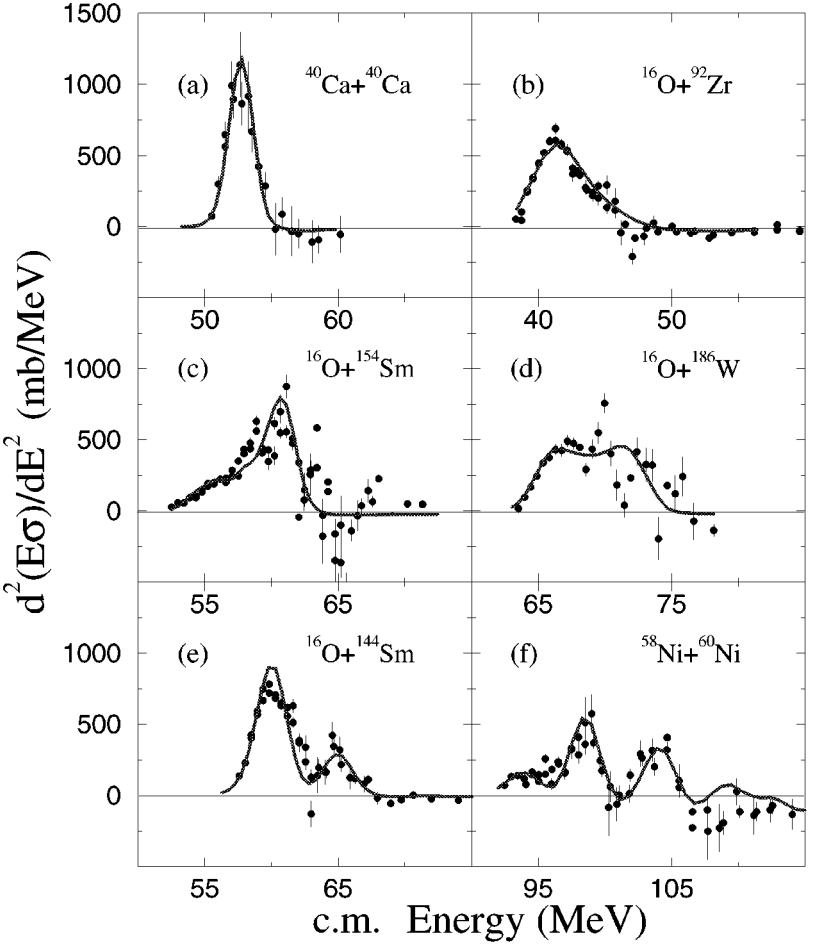


Figure 3 Experimental fusion barrier distributions for six different systems. (*a*) The case of ^{40}Ca + ^{40}Ca shows the prototype single-barrier result due to the double closed-shell nature of the target and projectile. (*b*) For ^{16}O + ^{92}Zr, an essentially continuous distribution is seen, even though the underlying theoretical structure is a set of discrete barriers due to phonon couplings. These are unresolved due to the relatively small value of Z_1Z_2. (*c*) ^{16}O + ^{154}Sm shows the expected continuous distribution for a deformed target. (*d*) The combination ^{16}O + ^{186}W again corresponds to a deformed target but with a negative β_4 as opposed to the positive value for ^{154}Sm. (*e*) For ^{16}O + ^{144}Sm, a second resolved bump occurs because of the coupling to target phonon states. (*f*) ^{58}Ni + ^{60}Ni shows a three-barrier structure owing to the complex surface vibrations induced in this reaction.

and the corresponding first and second derivatives,

$$\frac{1}{\pi R^2}\frac{d(E\sigma)}{dE} = \frac{1}{(1+e^x)} \qquad 5.$$

$$\frac{1}{\pi R^2}\frac{d^2(E\sigma)}{dE^2} = \frac{2\pi}{\hbar\omega}\frac{e^x}{(1+e^x)^2} \equiv G(E-B). \qquad 6.$$

These expressions will prove extremely useful. We note first, however, that the above analytic expression (Equation 6) fits rather well the single peak observed in the $^{40}Ca + {}^{40}Ca$ data.

We use the above results to obtain the classical result by taking the limit $\hbar \to 0$. This yields

$$E\sigma = 0 \quad \text{for} \quad E < B \quad \text{and} \quad E\sigma = \pi R^2(E-B) \quad \text{for} \quad E \geq B, \qquad 7.$$

from which we obtain

$$d(E\sigma)/dE = 0 \quad \text{for} \quad E < B \quad \text{and} \quad d(E\sigma)/dE = \pi R^2 \quad \text{for} \quad E \geq B \qquad 8.$$

and

$$\frac{1}{\pi R^2}\frac{d^2(E\sigma)}{dE^2} = \delta(E-B). \qquad 9.$$

Thus the function on the left-hand side of the above expression returns the barrier energy B. In the case where the total fusion cross section is given by an integral over a continuous distribution of barriers $D(B)$, i.e.

$$\sigma(E) = \int \sigma(E, B)\, D(B)\, dB, \qquad 10.$$

one then has

$$\frac{1}{\pi R^2}\frac{d^2(E\sigma)}{dE^2} = \int G(E-B)\, D(B)\, dB. \qquad 11.$$

In the classical limit, this gives the simple and striking result

$$\frac{1}{\pi R^2}\frac{d^2(E\sigma)}{dE^2} = D(E), \qquad 12.$$

that is, the quantity on the left-hand side of these equations directly reproduces the barrier distribution $D(E)$. In the quantum mechanical treatment, which includes tunnelling, the barrier distribution is smeared out by the function $G(E-B)$, as shown in Equation 11. Like $\delta(E-B)$, however, $G(E-B)$ is peaked at $E = B$, is symmetric about that point, and has unit area. The only difference is its finite width, FWHM $\approx 0.56\hbar\omega$ (7); thus the relation in Equation 11, which takes account of quantal tunnelling, gives the same information

as Equation 12 but smoothed over $0.56\bar{\hbar}\omega$, typically 2–3 MeV. If the intrinsic width of the distribution is large compared with this, the relevant information can still be extracted from the experimental data. Because the width of the overall distribution is determined by the strength of the couplings to excited states of the system, our main concern here is the study of heavy-ion fusion where the target and/or projectile have highly collective states or where strong transfer channels exist between the two.

To make some of these ideas more tangible, consider the case of a strongly deformed rotational target with axial and reflection symmetry. In a classical description of the appropriate dynamical variable (orientation), the total fusion cross section is given by (36, 37)

$$\sigma = \int_0^{\pi/2} \sigma(E, \Theta) \sin\Theta \, d\Theta, \tag{13.}$$

where Θ is the angle between the symmetry axis and the beam direction. Treating the barrier transmission classically yields

$$\frac{1}{\pi R^2} \frac{d^2(E\sigma)}{dE^2} = \int \delta(E - B) \sin\Theta \, d\Theta = \int \delta(E - B) \sin\Theta \, \frac{d\Theta}{dB} \, dB$$
$$= \sin\Theta \, \frac{d\Theta}{dB}, \tag{14.}$$

and so $D(B)\, dB = \sin\Theta \, d\Theta$, i.e. the normalized barrier distribution is given by the corresponding normalized solid angle. For a sharp nuclear surface with a quadrupole deformation (deformation parameter β_2), the barrier position and height may be parameterized as follows:

$$R(\Theta) \approx R_B + R_T \beta_2 \, Y_{20}(\Theta, 0) \tag{15.}$$

$$B(\Theta) \approx \frac{Z_1 Z_2 \, e^2}{4\pi\varepsilon_0} \frac{1}{R_B + R_T \beta_2 \, Y_{20}(\Theta, 0)}, \tag{16.}$$

where R_B is the average barrier radius and R_T is the average target radius. The range of barrier heights can be derived as

$$\Delta B \approx \frac{3}{2} \sqrt{\frac{5}{4\pi}} \, \frac{Z_1 Z_2 \, e^2}{4\pi\varepsilon_0} \frac{R_T}{R_B^2} \beta_2. \tag{17.}$$

Similar formulae hold for the vibrational case with the deformation parameter replaced by its rms value. The most important factor in the above equation, which one can vary by a suitable choice of target and projectile, is $Z_1 Z_2 \, \beta_2$. The first experiment (38) performed with these criteria in mind was the fusion of $^{16}\mathrm{O} + {}^{154}\mathrm{Sm}$, for which the above formula gives $\Delta B \approx 9$ MeV, a figure

significantly larger than $0.56\bar{h}\omega$. It is clear from Figure 3*c* that this rough calculation of the spread of the distribution agrees with the experimental range and also that the spread is sufficiently greater than the single-barrier width to show the structure appropriate for the deformed Sm target.

3.2 *Barrier Distributions for Non-deformed Nuclei*

The effect of vibrational states can be studied through the coupled-channels equations, although this often makes it difficult to visualize the main physical features of the reaction. Simple and elegant model problems were therefore used (28–30) to illustrate the main physical reasons why couplings to these degrees of freedom affect the fusion cross sections.

As a simple example, consider the one-dimensional problem where the target has a single excited state $|1\rangle$, which couples to the ground state $|0\rangle$ through a coupling form factor F. If we further assume that the energy of the excited state is negligible, then the coupled-channels equations are as follows:

$$\left[-\frac{\bar{h}^2}{2\mu}\frac{d^2}{dx^2} + V(x) - E\right]\phi_0 = F(x)\phi_1 \qquad 18.$$

$$\left[-\frac{\bar{h}^2}{2\mu}\frac{d^2}{dx^2} + V(x) - E\right]\phi_1 = F(x)\phi_0, \qquad 19.$$

where $V(x)$ is the internuclear potential, E is the energy, and ϕ_0 and ϕ_1 are the relative motion wavefunctions in the entrance and excited channels, respectively. These equations can be uncoupled by introducing $\chi_1 = \frac{1}{\sqrt{2}}(\phi_0 + \phi_1)$ and $\chi_2 = \frac{1}{\sqrt{2}}(\phi_0 - \phi_1)$ to obtain the following:

$$\left[-\frac{\bar{h}^2}{2\mu}\frac{d^2}{dx^2} + V(x) + F(x) - E\right]\chi_1 = 0 \qquad 20.$$

$$\left[-\frac{\bar{h}^2}{2\mu}\frac{d^2}{dx^2} + V(x) - F(x) - E\right]\chi_2 = 0. \qquad 21.$$

Since the entrance channel contains 50% of each of these eigenchannels (i.e. $|\langle\chi_{1,2}|\phi_0\rangle|^2 = \frac{1}{2}$), we obtain the total transmission coefficient, in terms of those corresponding to the uncoupled channels χ_1 and χ_2 (eigenchannels),

$$T = \frac{1}{2}[T(V+F) + T(V-F)]. \qquad 22.$$

In other words, the total transmission is a weighted sum of the transmission for the potentials $V + F$ and $V - F$. Thus, coupling to another channel effectively splits the potential barrier into two barriers. Transmission through the lower

barrier leads to an enhancement of the cross sections at energies below the uncoupled barrier.

The more general problem is discussed in Section 4.2., where it will be seen that rather than having a continuous distribution, the cross section is generally approximated by

$$\sigma(E) = \sum_{i=1}^{N} \sigma(E, B_i)\, w_i. \qquad 23.$$

In this case we have

$$\frac{1}{\pi R^2} \frac{d^2(E\sigma)}{dE^2} = \sum_{i=1}^{N} G(E - B_i)\, w_i. \qquad 24.$$

The number of barriers N is essentially equal to the number of strongly coupled channels (see Section 4). The weights w_i depend on the type of coupling involved and the reaction dynamics, so that the set of experimentally determined numbers w_i and B_i contains the essential physics of the problem.

3.3 *Representation of the Barrier Distribution*

Only in certain limits [adiabatic and isocentrifugal approximations (39–41)] is it possible to derive the barrier distribution analytically from the full quantum mechanical coupled-channels equations. Doing so outside these approximations is difficult, but both full theoretical calculations and experimental data clearly indicate that evaluating $d^2(E\sigma)/dE^2$ generates "distributions" with well-defined structures, giving significant insights into the role of the target and projectile structure in the fusion process. Conversely, the fusion barrier distribution can also give us information on nuclear structure in a form not previously suspected. To extract this information from the data, Rowley et al (7) proposed to define an experimental barrier distribution using Equation 2. That equation gives the same quantity as defined in Equations 11 and 24, apart from the normalization factor πR^2, which is not an experimental quantity—assumptions would have to be made to include it. Furthemore, the radius R is barrier dependent, a fact not taken into account in Equations 11 and 24, and one should not include a single constant value of this quantity in the expression. However, the area of the experimental barrier distribution should be roughly equal to the geometrical area for the reaction, i.e. πR^2, where R is an average of barrier radii. The barrier distribution must be evaluated by applying a point-difference formula to the data. As long as the comparison between theoretical and experimental distributions is made in the same way, neither of these two points causes any problems. Of course, in assessing the quality of any fit to experimental data, one should also compare cross sections. However,

the real physics of the problem is most visible in the distribution itself, which often contains a striking fingerprint of the couplings involved. Since a rigorous interpretation of $d^2(E\sigma)/dE^2$ as representing a set of effective barriers is not possible (except in the eigenchannel picture described in Section 3.2), henceforth the term barrier distribution will be used to mean the experimental barrier distribution defined by Equation 2.

4. REALISTIC MODELS

4.1 *Defining the Nuclear Potential*

The relative simplicity of the barrier distribution problem arises because the physics is rather insensitive to details of the monopole nuclear force but depends strongly on how that force is modified by structure effects. These two points will be demonstrated in the present section, where it will be shown that the monopole potential can be experimentally defined and the channel couplings derived simply from that same potential by letting the nuclear radius become an operator that depends on the static (rotational) or dynamical (vibrational) deformations of the nuclear surface.

The nuclear potential V_N may be parameterized in many different ways. The most fundamental basis for the interaction is probably the double-folding model (42). We should, however, consider just what we need to know about the real part of the potential in order to fit fusion data of the type we wish to discuss. The present problem, unlike the problem of elastic scattering with an "internal wave" passing through the nuclear interior (43), is one in which all internal flux is absorbed. Therefore, the only three properties of the problem that are of interest are the position R, height B, and curvature $\bar{h}\omega$ of the Coulomb barrier. For the systems of interest here, the barrier occurs at large distances where V_N is essentially exponential. Thus other details of the potential shape and parameters are not significant, so long as the correct barrier and its curvature are correctly reproduced. The parameters of the imaginary potential are also not significant, since the potential should simply be chosen to give total absorption in the nuclear interior. This may equivalently be replaced by an ingoing wave boundary condition. Thus the entire complex potential is defined by only two parameters. Changing the value of B simply shifts the entire structure of the barrier distribution by an appropriate amount and is, therefore, rather well determined experimentally. Changing the diffuseness a of the nuclear potential tends to have relatively little effect on the positions and weights of the barriers within the distribution, although it does determine the width of the smoothing function $G(E - B)$. It has, however, a significant effect on the cross sections at higher energies, well above the barrier region, since a small value of a gives a barrier position R that changes little with angular momentum,

whereas for a large diffuseness the barrier moves to substantially smaller R for high l values (44). From fits to the high-energy cross sections, large values of $a \approx 1$ fm are generally found (18, 34). It may be that these mock up an intrinsic energy dependence of the underlying bare potential or the effects of couplings that have been ignored (45). It is an interesting question whether the values of a determined from above-barrier fusion cross sections are consistent with those that may be determined from data at fusion barrier energies or below.

The most interesting dependence of the potential comes from the couplings to other channels, where the radius of the nucleus that possesses structure becomes a dynamical operator,

$$\hat{R}_N = R_0(1 + \hat{O}), \qquad 25.$$

where R_0 is the equivalent spherical nuclear radius. For rotational nuclei, the operator $\hat{O}$ is simply

$$\hat{O} = \beta_2 Y_{20}(\Theta, 0) + \beta_4 Y_{40}(\Theta, 0) + \cdots, \qquad 26.$$

and for quadrupole and octupole phonon states

$$\hat{O} = \beta_2^{\text{rms}}(A_2^\dagger + A_2) + \beta_3^{\text{rms}}(A_3^\dagger + A_3) + \cdots, \qquad 27.$$

where A and $A^\dagger$ are the corresponding phonon annihilation and creation operators. Thus, exactly the same potential that gives the central term is used to generate the couplings. In the first-order approximation, the coupling is given simply by

$$V_{\text{coup}} = -\frac{\partial V}{\partial r} R_0 \hat{O}, \qquad 28.$$

which stems from the fact that the nuclear potential is generally a function of $r - R_0$. In many cases, however, such an expansion is inadequate (and indeed unnecessary); the next subsection shows how coupling to all orders can be achieved.

4.2 *Realistic Calculations: How Many Channels, How Many Barriers?*

In the energy regime of interest, and for $Z_1 Z_2$ not so large as to encounter dissipative effects before the entrance-channel barrier is reached, the correct theoretical description of the fusion problem is the coupled-channels formalism. Calculations should include all the strongly coupled inelastic and transfer channels. This is easier said than done, since an important question is whether we understand the target and projectile structure sufficiently well to define these channels, and even if we do, can we perform a calculation in a large enough basis to obtain physically reasonable results?

One issue involved with such coupled-channels equations is whether they are solved exactly, in the sense of correctly treating the kinematics of the reaction. This means taking proper account of the excitation energies of the nuclear states as well as the "Coriolis" term in the centrifugal energy (essentially the difference of the centrifugal barrier in the entrance and exit channels). More significant, however, is whether they include the appropriate nuclear structure of the channels in sufficient detail. (Section 9 deals with this latter aspect of the problem.)

The treatment of complicated couplings presents a serious computational challenge if the reaction kinematics are to be treated exactly. The rest of this subsection is devoted to understanding this and to seeing how to reduce the size of the necessary calculations. More importantly, we show how this brings out the essential physics of the problem.

The coupled-channels equations for inelastic excitations may be written

$$(T_i + V(r) - (E - \epsilon_i))\psi_i(r) = \sum_j V_{ij}(r)\psi_j(r), \qquad 29.$$

where r is the target-projectile separation and T is the kinetic energy operator. The channel index i labels the excited state of the target (spin I_i) and the orbital angular momentum of the relative motion l_i, which are coupled to the initial entrance-channel value l_0, i.e. $I_i \otimes l_i = l_0$. Obviously the total angular momentum is conserved, and the equations must be solved for each value of l_0 separately. Several approximations to the above equations may be reasonable, depending on the system being considered.

The most important approximation is the so-called isocentrifugal approximation or orbital sudden approximation (39). The kinetic energy operator T_i depends on the channel angular momentum l_i through the centrifugal term. On average this will not differ much from l_0, so we may take $T_i \approx T_0$. Due to the large reduced mass of heavy-ion systems and the large barrier radii, this is a good approximation for all the systems considered here (40). In an exact coupled-channels treatment, each spin state I gives rise to $I + 1$ final channels with different angular momenta $l = l_0 - I, l_0 - I + 2, \ldots l_0 + I$. Ignoring the differences in the centrifugal potential (46) leads to a conserved quantum number that is essentially the projection of the nuclear spin along the recoil direction, i.e. the symmetry axis of the orbit. This is analogous to neglecting the Coriolis force in the particle-rotor or Nilsson models (47, 48), which leads to the conservation of the projection of the nucleon angular momentum along the nuclear symmetry axis. Transforming to this strong-coupling basis, the channels can be labeled by (I, K), where K is the spin projection along the recoil axis. Explicitly we have $(I, K) = (I, 0),\ (I, 1) \ldots (I, I)$. However, since the entrance channel possesses only $K = 0$, and since K is conserved, only the single channel $(I, 0)$ remains and thus there is only one coupled equation

for each nuclear state included. The isocentrifugal approximation leads to an enormous reduction in the dimensionality of the problem and is applicable to all inelastic couplings of any complexity, whether rotational or vibrational. It is essentially the exploitation of this result that has led to the possibility of including the complicated couplings demanded by the new, precise data. The above result also supports the previously mentioned expectation of one barrier for each nuclear state included. It is possible to take this discussion further and analytically prove a one-to-one correspondence between channels and barriers by making two further approximations: (*a*) We may neglect the excitation energies ϵ_i. This "adiabatic" approximation is reasonable only for strongly deformed nuclei. (*b*) Finally, all the coupling form factors can be assumed to have the same radial dependence, i.e. $V_{ij} = \Gamma_{ij} F(r)$.

With these approximations, Equation 29 becomes

$$(T_0 + V - E)\psi_i(r) = F(r) \sum_j \Gamma_{ij} \psi_j(r), \tag{30.}$$

and we see that this equation may be diagonalized with the unitary transformation U, which diagonalizes Γ:

$$U^{\dagger}_{\alpha i} \Gamma_{ij} U_{j\beta} = \lambda_\alpha \delta_{\alpha\beta}. \tag{31.}$$

This gives

$$(T_0 + V(r) - \lambda_\alpha F(r) - E)\Psi_\alpha(r) = 0, \tag{32.}$$

where the "eigenchannel" wavefunction is a linear superposition of the physical channels,

$$\Psi_\alpha = U_{\alpha i} \psi_i. \tag{33.}$$

In other words, the problem reduces to one of solving a set of uncoupled equations for a series of different potentials $V - \lambda_\alpha F$. The fusion cross section has then the form of Equation 23, where the individual cross sections are calculated in these potentials. The weights are given by

$$w_\alpha = |U_{0,\alpha}|^2 = |\langle \psi_0 | \Psi_\alpha \rangle|^2, \tag{34.}$$

i.e. the fraction of eigenchannel α contained in the entrance-channel configuration $|\psi_0\rangle$.

Consider again the case of the even-even deformed rotor. Its ground state wavefunction (spin 0) possesses components corresponding to all orientations. Conversely, the wavefunction corresponding to a fixed orientation (delta-function wavepacket) must possess components corresponding to all angular momenta. Therefore, in the language of quantum mechanics, the previous

classical description must be intimately related to the excitations of the rotational states of the deformed target. In this case, one may show (25, 49) that the above sum (Equation 23) is the Gauss-Legendre approximation to the integral (Equation 13) with a number of points equal to the number of target states (hence barriers) included. Thus, when all states of the rotational band ($I \to \infty$) are included, the quantal and classical descriptions of the nuclear orientation give identical results. (It is interesting to note that the adiabatic and isocentrifugal approximations are implicit in the above classical description of this problem.)

These results are very significant because one can prove analytically that the fusion cross section may be expressed as a sum over a set of eigenchannels. But what happens if the approximations we have made are no longer valid? For heavy nuclei the isocentrifugal approximation is always reasonable, but the adiabatic one $\epsilon \approx 0$ often is not. For example, quadrupole and octupole vibrational modes (phonon states) usually have large excitation energies (>1 MeV), and then it is necessary to solve the reduced set of coupled-channels equations numerically or to invoke further approximations (30, 50, 51).

Neutron-transfer channels pose additional problems if an analytic (eigenbarrier) solution is sought, since strictly speaking the relevant couplings should be expressed as nonlocal potentials. However, as discussed in Section 3.3, it is not necessary to calculate a set of eigenbarriers (a barrier distribution in the strict sense). Instead, an exact (as far as possible) calculation can be performed, and the "barrier structure" can be determined by evaluating $d^2(E\sigma)/dE^2$. Neutron-transfer channels are of special interest in that, if the transferred neutron is more strongly bound after transfer, the reaction is not dissipative but exothermic and increases the kinetic energy of relative motion. We shall see later that in certain cases this process can be highly collective (52, 53) and may be thought of as a neutron flow.

Let us show explicitly, in the context of vibrational couplings, how the number of channels can be drastically reduced. Consider the problem of a system containing many phonon states. Specifically, take two types of phonon, i.e. quadrupole 2^+ and octupole 3^- with zero-, one-, and all possible two-phonon excitations. The above system then corresponds (54) to the daunting number of 53 coupled channels! The no-Coriolis approximation immediately reduces this to 15. The fact that all the spin states arising from $2^+ \otimes 2^+ (= 0, 2, 4)$ are degenerate allows us to make a single linear superposition of these states, to which the one-phonon state couples. This and the analogous reduction for the other two-phonon modes further reduces the number of coupled equations to a mere six. This yields, therefore, an eminently feasible calculation. Of course, in a real nucleus, the two-quadrupole–phonon states are actually split in energy, and the above reduction would be an approximation, albeit a good one so long

as the splitting is relatively small. We see, therefore, that the corresponding barrier distribution, though not apparent a priori from the original problem of 53 coupled channels, should have a relatively simple structure, which can be extracted from sufficiently precise experimental data.

Another interesting case is when both nuclei have single-phonon states and the multiphonon states correspond to mutual excitation of target and projectile. Similar considerations again apply (55).

The program CCFUS has been extensively used to fit fusion data (50, 51). It is very simple to apply, since the barrier heights and weights are obtained analytically through an approximation that makes a complete diagonalization possible even for finite excitation energies. This approximation is to replace the coupling form factor by a constant value evaluated at the barrier. [An essentially equivalent formalism was introduced (28) wherein the excitation energies were multiplied by the same radial form factor as the couplings.] Other variants take into account the fact that the different barriers occur at different radii (56).

Recent, detailed data show that if the relevant excitation energies are large, these procedures generally do not give sufficiently good results. The major reason for this actually lies in the long-range Coulomb excitation of the system. If the excitation energies are large, the system has a long time to respond to the noncentral Coulomb forces and arrives in the barrier region in a configuration different from that of the pure entrance channel. Since we are often interested in systems with a large Z_1Z_2, Coulomb excitation will be important, and thus couplings to states with large excitation energies will always require a proper coupled-channels calculation. We have shown that in many cases the basis can be very significantly reduced to make this feasible.

4.3 *Calculations with Realistic Couplings*

Many earlier calculations, although treating the kinematics in full detail, included couplings only to first order. This is clearly insufficient for coupling to a rotational band. However, the power of the new, precise data and their visualization through the experimental barrier distribution showed the inadequacy of first-order couplings for many cases involving phonon excitations. This inadequacy can be appreciated from Figures 3*e* and 3*f*, which show the data for $^{16}O + {}^{144}Sm$ and $^{58}Ni + {}^{60}Ni$. The former (57) possesses a simple second bump principally due to coupling to the 3^- octupole phonon state and can be reasonably well reproduced by a first-order calculation. However, the case of $^{58}Ni + {}^{60}Ni$ (34) requires all-order couplings up to the mutual double excitation of the quadrupole phonon states of the target and projectile in order to fit the data well. Such complexity had not been previously imagined and has many important consequences. The evaluation of the coupling form factors in

such a case can be achieved by some simple tricks of matrix algebra that are greatly preferable to making an expansion in powers of the coupling strength (54, 55, 58). For the rotational case, full details are available elsewhere (49). Here the results for the general problem are schematically derived.

In this general case, the potential becomes a function of the operator $\hat{O}$, i.e. $V_N = V_N(\hat{O})$, and the channel couplings we need to evaluate are

$$V_{ij} = \langle \phi_i | V_N(\hat{O}) | \phi_j \rangle, \qquad 35.$$

where ϕ_i are the nuclear wavefunctions in the physical channels *i*. The simpler matrix

$$M_{ij} = \langle \phi_i | \hat{O} | \phi_j \rangle \qquad 36.$$

can be diagonalized by a unitary transformation *U*, i.e.

$$\tilde{M}_{\alpha\beta} = U^{\dagger}_{\alpha i} M_{ij} U_{j\beta} = \langle \alpha | \hat{O} | \beta \rangle = \lambda_\alpha \delta_\alpha. \qquad 37.$$

Of course $|\alpha\rangle = U_{\alpha i} |\phi_i\rangle$ are the same eigenchannels previously introduced, since if they are eigenstates of $\hat{O}$ they are also eigenstates of $V_N = V_N(\hat{O})$. Although they are not eigenchannels of the full problem unless we make the adiabatic approximation, we may still exploit them to obtain the physical channel couplings, using

$$V_{ij} = \langle \phi_i | V_N(\hat{O}) | \phi_j \rangle = U_{i\alpha} V_N(\lambda_\alpha) U^{\dagger}_{\alpha j}. \qquad 38.$$

In this way, once the physical space has been defined, the coupling potential may be evaluated to all orders in that space. Such a procedure is easy to apply and is essential, for example, in the correct description of the barrier distribution in the rotational problem via a quantum mechanical calculation. Similar comments apply to vibrational systems.

Much recent work has exploited the Interacting Boson Model (IBM) in fitting fusion barrier distributions (59). The only essential difference in this approach is that the generalized operators of the above formalism become the boson creation and annhilation operators, as opposed to the phonon creation and annhilation operators or the "macroscopic" operator Y_{20}.

The above discussion is essentially limited to inelastic couplings. It is shown below, however, that certain systems appear to exhibit strong effects from nucleon-transfer channels. The main effects come from neutron transfers, since these are not inhibited by the Coulomb barrier. For relatively weak couplings, such channels can be modelled by fictitious inelastic states, but the main interest lies in strong effects of multinucleon transfers because these have the largest effect on the reaction dynamics. The reduction of the lowest barrier is frequently

so dramatic (see e.g. the discussion of $^{40}Ca + {}^{90,96}Zr$) that it becomes appropriate to think of the formation of a neutron neck (60), and then more detailed calculations, or the construction of some appropriate macroscopic model, may be required—especially when the inelastic couplings remain strong. Remaining within the framework of complete coupled-channels calculations, a code such as FRESCO (61) should prove useful.

The basic philosophy of this code is that local (inelastic) couplings are treated to all orders, but that transfers are treated iteratively, often using a Padé acceleration method to achieve convergence. It can also treat inelastic couplings iteratively, and if they are weak then this is generally faster. In practice, one selects the strongly coupled channels for all-orders exact solutions, and treats other inelastic and transfer channels in an iterative manner. The drawback of such exact treatments is the sheer magnitude of such calculations for complex systems. It may also be unclear whether transfers to excited states, or other effects such as pair transfer, are important; that is, the physical input necessary may have to be learned from the experimental results. A full analysis, as was performed for $^{16}O + {}^{208}Pb$ (62), of some of the systems in question would be very time-consuming, but several systems may have reached the stage where such an investment would be justified.

5. EXPERIMENTAL METHODS

5.1 *Experimental Definition of Fusion*

In contrast with the conceptual definition of fusion in terms of a barrier-passing picture, experimentally, fusion must be defined in terms of the nuclei formed in the reaction or the products of their later decay, since these are the quantities that are measureable. They are not determined solely by the fusion potential, but are controlled by the subsequent evolution of the system. As the two nuclei captured in the fusion potential move closer together, excitation of very many internal degrees of freedom occurs. Kinetic energy is dissipated, a neck forms, and the two nuclei come to thermal equilibrium. At this point they can be considered as a single elongated and necked nucleus called a dinucleus. Despite being captured in the fusion potential, the dinucleus may never form a compound system with its equilibrium deformation. The evolution of the system after capture can be described largely in terms of its shape and is governed by the friction and inertia tensors and the potential. Since all but the shape degrees of freedom are assumed to be in equilibrium, we call it the fission potential.

For light dinuclei, the shape evolution in the fission potential results in the formation of a compound nucleus at its equilibrium deformation, which subsequently decays by evaporation, leaving an evaporation residue. The direct or

indirect detection of evaporation residues is indisputable experimental evidence of fusion. For heavier (more fissile) systems, fission becomes a likely decay mode but is still associated with decay of an equilibrated compound nucleus. For such nuclear collisions, the definition of fusion in terms of the fusion potential, resulting in capture, is equivalent to the definition in terms of the final products.

It is known experimentally that for reactions involving still heavier nuclei, e.g. $^{40}\mathrm{Ca} + {}^{238}\mathrm{U}$, the characteristics of the fission fragments are not consistent with statistical decay of a compound nucleus (63, 64), although the kinetic energy of the projectile is completely damped. Such a process is called quasi-fission. It is interpreted (65) as capture in the fusion potential (i.e. passage inside the fusion barrier), with formation of a dinucleus. However, the evolution to the fission potential, and the corresponding reorganization of the composite nucleus results in the potential ultimately becoming repulsive; thus the nucleus splits after undergoing substantial or almost complete mass equilibration (63).

If experimental data are compared with a model containing only the fusion potential, then the quasi-fission should be included in the fusion yield (66). If a model also aims to describe the subsequent evolution over the fission potential, and thus the true compound nucleus formation cross section, then quasi-fission should not be included in the fusion cross sections. This review discusses mainly the former case.

5.2 *Extraction of Experimental Barrier Distributions*

Experimental cross sections can be determined only at a finite number of energies and are thus discrete. Theoretical cross sections can sometimes be obtained analytically, permitting exact determination of the barrier distribution. Realistic calculations, however, seldom permit an analytic solution. In such cases, the cross sections can in principle be calculated in very small energy steps. However, since the calculations are time-consuming, in practice they are done with larger energy intervals. From experimental or calculated cross sections with finite energy intervals, the second derivative of Eσ can be obtained approximately from the successive determination of the first derivative using a point difference formula. At energy $(E_1 + 2E_2 + E_3)/4$, it is given by

$$\frac{d^2(E\sigma)}{dE^2} = 2\left(\frac{(E\sigma)_3 - (E\sigma)_2}{E_3 - E_2} - \frac{(E\sigma)_2 - (E\sigma)_1}{E_2 - E_1}\right)\left(\frac{1}{E_3 - E_1}\right), \qquad 39.$$

where $(E\sigma)_i$ are evaluated at energies E_i. Because this method of obtaining $d^2(E\sigma)/dE^2$ inherently introduces some smoothing, it is essential to treat both experimental and calculated cross sections in the same way when comparing experiment and theory. For data with equal energy steps,

$\Delta E = (E_2 - E_1) = (E_3 - E_2)$, the above expression reduces to

$$\frac{d^2(E\sigma)}{dE^2} = \left(\frac{(E\sigma)_3 - 2(E\sigma)_2 + (E\sigma)_1}{\Delta E^2} \right). \quad 40.$$

Then the statistical error δ_c associated with the second derivative at energy E is approximately given by

$$\delta_c \simeq \left(\frac{E}{\Delta E^2} \right) \left[(\delta\sigma)_1^2 + 4(\delta\sigma)_2^2 + (\delta\sigma)_3^2 \right]^{1/2}, \quad 41.$$

where the $(\delta\sigma)_i$ are the absolute cross section uncertainties. In a measurement, $\delta\sigma/\sigma$ generally reaches a lower limit as the cross section rises, and so beyond this point, δ_c is proportional to the value of σ and increases with increasing beam energy. The barrier distribution is therefore well defined at the lowest energies, where the cross sections are small (even though $\delta\sigma/\sigma$ may be relatively large), but it becomes increasingly poorly defined as the energy increases.

The value of δ_c is inversely proportional to ΔE^2 and can therefore be reduced by using larger energy intervals. Of course, large intervals smooth the extracted distribution, resulting in the damping of features in the distribution whose energy scale is smaller than the step size. As the barrier distribution is already physically smoothed by $0.56\hbar\omega$ (typically ≈2–3 MeV) because of tunnelling, the use of $\Delta E \approx 2$ MeV does not result in significant additional smoothing.

The curves in Figure 4 show barrier distributions for a hypothetical system, evaluated using the point-difference formula, from cross sections calculated at 1-MeV intervals with a coupled-channels code. The smoothing that results from calculating the barrier distribution with energy steps of $\Delta E = 2$ MeV and 3 MeV (Figure 4*a*, *b*) is compared with the almost unsmoothed distribution determined with $\Delta E = 1$ MeV (*dashed lines*). Sharp structures are only significantly attenuated for $\Delta E = 3$ MeV. Experimental data have been simulated by randomly scattering these calculated cross sections, with a Gaussian distribution of standard deviation 0.5%. The reduction in scatter of the experimental barrier distribution at the higher energies is clear as ΔE is changed from 2 to 3 MeV (compare panels *a* and *b*), but the attenuation of structure is also apparent. Typically, the analysis of real experimental data uses a value of $\Delta E = 2$ MeV as a compromise between resolving structure and reducing scatter. If the data and calculations indicated that the barrier distribution had no sharp structures, a larger energy step could well be justified.

Because a 2-MeV energy step is used to determine the barrier distributions, the same energy interval between cross section measurements may seem ideal. This is not the case, as a large deviation of one point (by chance or by human error) would also affect the two neighboring points, resulting in an apparent feature of similar width to a natural barrier. To avoid this possibility, it is

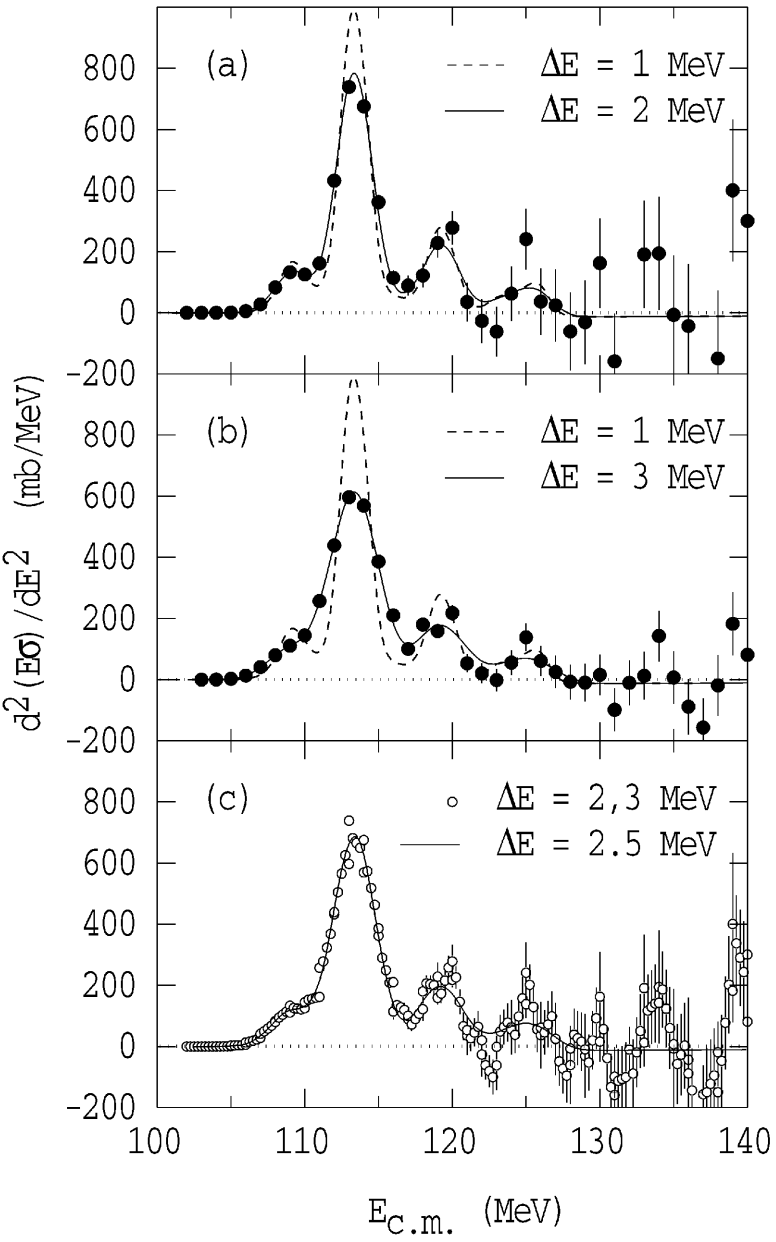


Figure 4 Fusion barrier distributions determined using the point difference formula on a calculated excitation function (*curves*) and on simulated data points randomly scattered around the calculation, for different energy steps.

important to have cross section data with significantly smaller energy steps, so that several independent measures of D_{exp} exist within the energy interval ΔE.

The barrier distribution could be evaluated using different energy intervals above and below a given point (e.g. intervals of $+2$ MeV, -3 MeV, and $+3$ MeV, -2 MeV, as well as ± 2 MeV and ± 3 MeV), resulting in more values in a given energy interval. However, these are not independent; in this example, a single deviant cross section will affect 12 values of the distribution! This can be seen in Figure 4*c*, where apparently significant structures at 134 and 139 MeV are generated by one or two points randomly thrown more than one standard deviation from the calculation. The parent distribution (smooth curve), evaluated for a 2.5 MeV energy step, shows no structure at these energies. When the energy step in the cross section data is considerably less than 1 MeV, the density of data is high enough that any features are well defined using the normal procedure; so taking multiple energy steps is not recommended.

Other methods to determine the barrier distribution from experimental data are possible [e.g. (67)], but these have not yet been widely discussed in the literature. As a general principle, it is probably preferable to obtain data with lesser statistics at more energies, than with better statistics at fewer energies.

Evaluation of the barrier distribution from a measured fusion excitation function serves a very useful purpose, even though theoretically it can be argued that the barrier distribution concept is only valid under certain approximations. The scatter of the data points and the size of the error bars show whether the data are of sufficient quality to allow meaningful conclusions to be drawn, for instance, in distinguishing between two model calculations. It is noteworthy that inevitably the absolute error bars on cross sections increase with energy; one must take this into account if one is making a χ^2 fit to cross section data alone. Small differences in the calculated barrier distribution below the average barrier will affect the χ^2 very severely, while a gross change above the average barrier may make a much smaller change to χ^2. This could result in misleading conclusions, which can be most easily prevented by also comparing the barrier distributions.

5.3 *Requirement for Precise Data*

The second derivative of experimental data over the relevant energy range is typically determined from small differences between large cross sections. Thus, fluctuations in the cross sections due to experimental uncertainties must be smaller than the changes due to the curvature. The quality of experimental data can be defined in terms of the precision (related to repeatability) and the accuracy (related to the degree of systematic uncertainties or errors). Precision is more important, since it is differences of neighboring points that define the barrier distribution.

Ideally, to obtain the smallest uncertainties, all fusion products or their decay products would be measured with 100% efficiency. Realistically, a detector with a limited solid angle is used, only allowing measurement of $d\sigma/d\Omega$. Measurement of angular distributions is also necessary to obtain the integrated cross section σ. Angular distributions usually change little over the energy range used to obtain the curvature, so uncertainties in σ will come largely from uncertainty in $d\sigma/d\Omega$, which arises from statistical uncertainty due to the number of counts and from uncertainty in the angle of the detector. The latter can be minimized by choosing an angle Θ where $d\sigma/d\Omega$ is not strongly dependent on Θ. A precision of $\sim$1% is typical in measurements.

The presence of systematic errors, either with a slow beam-energy dependence or independent of beam energy, results in a direct scaling of the barrier distribution. Unless the error is large (>5%), the interpretation of the barrier distribution will not be seriously affected, since random errors in D_{exp} usually exceed 5% at energies above the mean barrier energy. Systematic errors can be important, however, in studying other aspects of the fusion process.

Determination of the curvature requires knowledge of the beam energy. The change in beam energy corresponding to a 1% variation in cross section at above-barrier energies is typically 50 keV. In order to avoid introducing additional uncertainty, the beam-energy step should be known much better than this. A systematic error in the beam energy will cause the mean fusion barrier to be incorrect but will not significantly affect the barrier distribution. Leigh et al (18) describe measures taken at the Australian National University (ANU) to minimize errors in beam energy.

5.4 *Measurement of Evaporation Residues*

Evaporation residue cross sections can be determined by direct detection of the evaporation residues themselves or by the detection of the radiation emitted in their deexcitation. In the direct detection methods, the evaporation residues, which are forward peaked, must be physically separated from the direct beam and the intense flux of elastically scattered beam particles. Separation can be achieved using electric and/or magnetic fields, and a variety of devices employing such techniques have been described (68). However, not all these devices are suitable for the high-precision measurements required to obtain fusion barrier distributions.

Separation of evaporation residues from the beam particles at 0^0 and at small angles has been achieved by applying an electrostatic field perpendicular to the direction of the particles, placed between the target and the detection system. Due to the difference in electrical rigidity (energy per ion charge state) usually existing between evaporation residues and beam, the two kinds of particles are separated (typically by a few centimeters) downstream of the electric field. Such

an electrostatic beam deflector (69) has been used in the Legnaro experiments on heavy-ion fusion cited below, where an energy and time-of-flight telescope, based on micro-channel plates and Silicon surface barrier detectors, detects and identifies the evaporation residues. The electrostatic filter can reduce the rate of beam and beam-like particles entering the telescope by a factor 10^7–10^8.

If the momentum of the compound nucleus is small, as is the case in mass-asymmetric reactions, residue energies can differ by a factor of 4 or more due to particle evaporation. This produces a very large spatial distribution of evaporation residues when traversing only an electrostatic field, making it impossible to transmit all the residues through the deflector. In such cases, the ions may be transmitted through a region of crossed electric and magnetic fields (a Wien filter) or through separated electric and magnetic fields. By selecting a particular ratio of the electric to magnetic field strengths, only particles in a certain velocity range are transmitted. The beam particles suffer more deflection than the evaporation residues and are physically stopped before entering the detector behind the filter.

The compact Wien filter (70) at the ANU transmits 100% of evaporation residues passing the solid angle–defining aperture before the filter, at the expense of a small solid angle. The residues are detected in a multiwire proportional counter behind the filter and are identified by their time of flight and their energy loss in the detector. This device has allowed very precise and accurate measurement of fusion cross sections for mass-asymmetric reactions.

In contrast, the velocity filter SHIP (71) at the GSI laboratory has a large solid angle, at the expense of having to calculate the transmission efficiency. This device has allowed the measurement of small evaporation-residue cross sections for near-symmetric fusion reactions (9), as well as the discovery of the heaviest elements.

Detection of the decay products of the evaporation residues allows separation in space, and often in time, of the beam particles and the detectors. Because of its full detection efficiency and characteristic signature, detection of alpha-particle decay has been successfully used in the determination of fusion barrier distributions for particular reactions (72) in which alpha decay is the dominant or exclusive ground-state decay mode. Measurements of gamma rays (73, 74) and delayed X rays (75), as well as evaporated neutrons (76), have also been used to measure fusion cross sections, although high precision may be difficult to achieve, partly due to uncertainties in feeding of the decaying states, in branching ratios, and in absolute detection efficiency.

5.5 *Measurement of Fission Fragments*

Fission decay of the fused system is significant for heavy compound nuclei ($A \geq 170$), if not at the fusion barrier energy, then at higher energies. For

compound nuclei with A $\geq$ 220, fission is the dominant decay mode at all beam energies. Measurements of the fission cross sections are thus necessary in many reactions, as they either complement the larger evaporation residue cross sections or are themselves the dominant fusion products.

Fission fragments are easily identified, either by their kinematic coincidence in two detectors (66) or by single detectors using energy and/or energy-loss and/or time-of-flight information (72, 77). They are spread to all angles, giving good separation from beam particles, but their angular distribution can change quite rapidly with beam energy, requiring careful placement of the detector (77) or measurement of the angular distribution at many (or all) beam energies (72, 66).

For reactions on fissile target nuclei (actinides), transfer reactions also lead to fission, requiring the identification and elimination of this component. This has been achieved by the measurement of the folding angle of the two fragments, but is most cleanly accomplished by the kinematic coincidence method (78), using the derived velocity components of the fissioning nuclei (79). As is discussed in Sections 11.2 and 11.3, the properties of the fission fragments themselves carry useful information on the reaction process.

6. ROTATIONAL NUCLEI—TESTING THE CONCEPT

Since the existence of a distribution of barriers is theoretically well established for reactions with deformed nuclei, these reactions were the obvious choice to test the concept of extracting an experimental barrier distribution.

Figure 5*a* shows the barrier distribution for the $^{16}O + {}^{154}Sm$ reaction, extracted from cross sections (16) that had a quoted experimental uncertainty of $\pm 10\%$ (a level typical for fusion data prior to the realization of the value of greater precision). Such data yield poorly defined barrier distributions, which, though clearly not consistent with a single barrier, can be equally well reproduced by different models predicting barrier distributions with very different shapes. This is illustrated by the good agreement of the data with both the continuous and dot–dashed lines in Figure 5*a*, which represent, respectively, the expectations for a deformed target and a Gaussian distribution of barrier heights. A meaningful test of the concept of barrier distributions clearly needed the fusion excitation function to be measured to much higher levels of precision.

This prompted experiments at the ANU, where the first high-precision (uncertainty $\pm 1\%$) measurements of fusion excitation functions were made with the aim of extracting fusion barrier distributions. The first reaction measured, $^{16}O + {}^{154}Sm$ (38), was chosen because ^{16}O was expected to be inert, and ^{154}Sm has a well-known static deformation. Figures 5*b* and 6*b* show the fusion

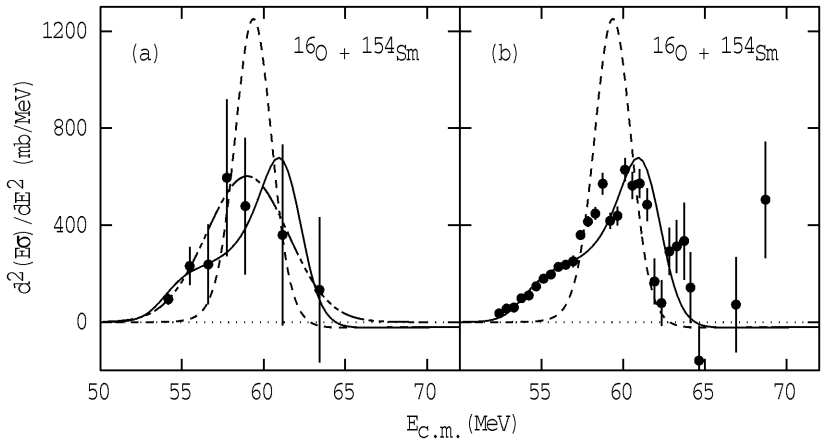


Figure 5 Precise excitation functions are necessary to obtain meaningful barrier distributions. Two barrier distribution are shown, extracted from data with uncertainties of (*a*) 10% and (*b*) 1%. The data in panel (*a*) are unable to distinguish between the two calculations shown by the *dot-dashed* and *solid curves*, although they clearly eliminate a single barrier (*short-dashed curve*). The data are taken from Refs. 16, 18.

barrier distribution for this reaction, extracted from these data. The advantages of high-precision measurements are clear in defining the experimental distribution at the level where it can now distinguish between the two different calculations.

One of the most interesting results of this first measurement was the sensitivity of the barrier distribution not only to the quadrupole deformation but also to the hexadecapole deformation (38). This became even clearer when a comparison of the experimental barrier distributions for the $^{16}O + ^{154}Sm$ and $^{16}O + ^{186}W$ reactions became possible.

Of the rare-earth nuclei, ^{154}Sm is predicted to have among the largest positive hexadecapole moments ($\beta_4 \sim +0.06$), while ^{186}W has among the most negative ($\beta_4 \sim -0.06$). The upper and lower panels of Figure 6 show the excitation functions (38, 80) and barrier distributions for $^{16}O + ^{154}Sm$ and ^{186}W. It is clear that the shape of the barrier distribution is very different for these two reactions. Attempts to fit the data by inclusion of only a static quadrupole deformation (*long-dashed lines*) appeared to give good agreement with the experimental cross sections, but the values of β_2 giving the best fits to the data were larger than those deduced from Coulomb excitation measurements (81). Moreover, comparison of the measured and calculated barrier distributions showed significant disagreements, which are different for the two reactions, supporting the expected sensitivity to the hexadecapole deformation.

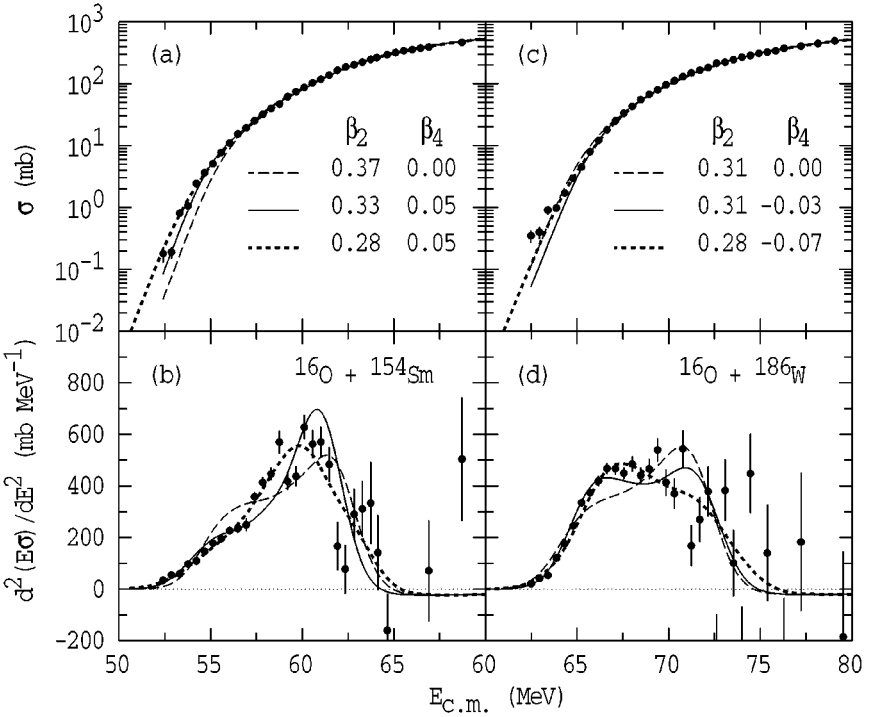


Figure 6 Comparison of the measured fusion excitation functions and the extracted barrier distributions for the indicated reactions. The *dashed lines* show the calculations including only quadrupole deformations of ^{154}Sm and ^{186}W, the *solid lines* include hexadecapole deformations while the *short-dashed lines* include the expected small effects of phonon and transfer couplings. The difference in the shapes of the barrier distributions is essentially due to the different sign of the hexadecapole deformations for the two nuclei. From Ref. 18.

Fitting the data including the hexadecapole degree of freedom gave the results indicated by the solid lines, which agree much better with the experimental data. This showed conclusively that the dramatic differences in the shapes of the experimental barrier distributions for the two systems are mainly due to the differences in the sign of the β_4. Minor discrepancies between the calculated and experimental distributions are still present. Inclusion of small phonon and transfer couplings, the presence of which is expected, allows even better reproduction of the data (18) (*short-dashed lines*), the best fit requiring deformation parameters very close to those determined from Coulomb excitation studies (81).

The results of these measurements exceeded all expectations, showing that fusion excitation functions are very sensitive not only to the quadrupole

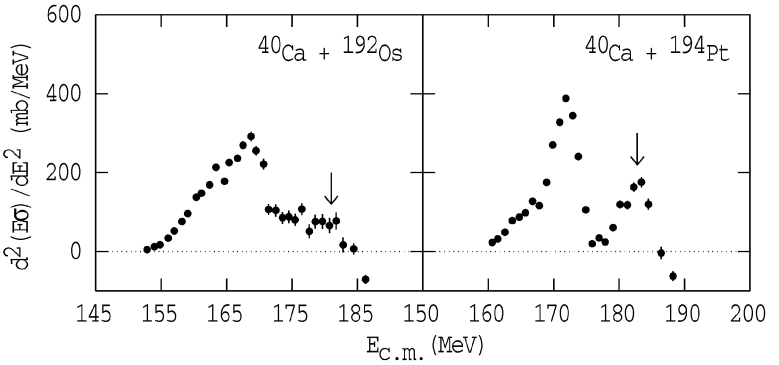


Figure 7 Experimental fusion barrier distributions for the ^{40}Ca + ^{192}Os, ^{194}Pt reactions have a different structure due to the oblate and prolate shapes of the target nuclei; the *arrow* indicates the feature arising as a result of coupling to the 3^- state in ^{40}Ca (see Section 9). The data are from Ref. 77.

deformation but also to small hexadecapole deformations, the signs of which are difficult to determine unambiguously by other techniques. The results also showed that, for these statically deformed nuclei, the function $d^2(E\sigma)/dE^2$ does indeed return a sensible representation of the barrier distribution expected to be present.

Measurement of the fusion barrier distributions for the deformed targets ^{192}Os and ^{194}Pt was subsequently suggested (82), since these are expected to have quadrupole moments with opposite sign, and the fusion barrier distributions for oblate and prolate nuclei are dramatically different. Measurements were performed at the University of Washington (77) for the reactions ^{40}Ca + ^{192}Os, ^{194}Pt detecting fission as the signal that fusion had occurred. The large charge product in these reactions results in a broad barrier distribution, which was expected to allow a very clear picture of the nuclear shapes. Interestingly, the barrier distributions also showed significant features attributed to excitation of the 3^- state in ^{40}Ca, and positive Q-value transfer reactions, which complicate the interpretation. Nevertheless, the barrier distributions (see Figure 7) exhibit a significant difference in the shape of the main peak, which could be explained by the prolate/oblate shape transition.

7. REACTIONS INVOLVING VIBRATIONAL NUCLEI

The precision measurements discussed above for reactions with rotational nuclei provided a quantitative test of the barrier distribution concept. For these

cases, the barrier distribution can be modeled classically. It was then crucial to test whether the concept would still be applicable where the existence of a distribution of barriers is not classically apparent. Furthermore, as discussed in Sections 3 and 4, analytic derivation of the barrier distributions can only be done under the adiabatic ($\epsilon = 0$) and isocentrifugal limits; thus it was of interest to see if measured barrier distributions would show the predicted effects of couplings to vibrational states with typical excitation energies of a few MeV.

7.1 *The* ^{16}O, ^{28}Si + ^{144}Sm *Reactions*

The measured (57) excitation function for the ^{16}O + ^{144}Sm reaction and the extracted barrier distribution are shown in Figure 8*a*, *b*. Based on earlier studies

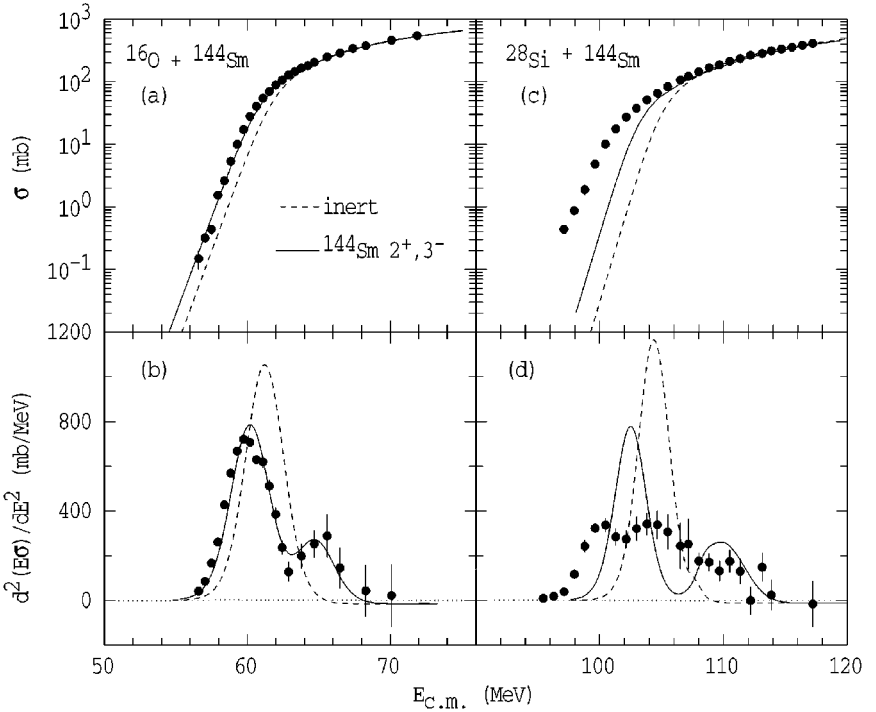


Figure 8 Comparison of the measured excitation functions and extracted barrier distributions for ^{16}O + ^{144}Sm (*left panels*) and ^{28}Si + ^{144}Sm (*right panels*) reactions. Couplings to the 2^+ and 3^- states of ^{144}Sm are adequate to describe the ^{16}O data as shown by the *solid lines* in the left panels. These couplings, however, are not sufficient for the ^{28}Si data, which indicate that additional couplings to states in ^{28}Si must be included. Data are from Ref. 18, 84.

(83), this reaction might have been expected to show a distribution consistent with a single barrier, although some small effects could have been expected from couplings to the inelastic channels associated with the semi-magic ^{144}Sm. The experimental barrier distribution, however, exhibited a distinct two-peaked structure, with the main strength around 60 MeV, and a peak with smaller strength near 65 MeV.

Calculations assuming a single barrier (no coupling), shown by the *dashed lines*, are clearly inconsistent with the excitation function and the barrier distribution. Whereas it may seem plausible to obtain a fit to the excitation function by a simple renormalization of the bare potential, such a procedure is not able to reproduce the excitation function accurately enough, a fact more readily apparent from the barrier distribution plot. As the *solid lines* in Figure 8 show, an excellent representation of the excitation function and the double-peaked structure of the barrier distribution was obtained by including couplings to the 2^+ and 3^- states of ^{144}Sm, the latter having the dominant effect. The clear qualitative agreement, further supported by observation of similar double-peaked structure in the barrier distribution for the $^{17}\mathrm{O} + {}^{144}\mathrm{Sm}$ reaction, provided an unambiguous experimental signature of the barrier-splitting effect resulting from coupling from the ground state to a vibrational state or states. These results also lent credibility to the barrier distribution picture even in the case where the excitation energies are not very close to zero, as in the eigenchannel approach. They showed that the barrier distribution could be used as a tool to understand more complex vibrational couplings.

The simple structure of the barrier distribution for $^{16}\mathrm{O} + {}^{144}\mathrm{Sm}$ changes drastically (84) when ^{28}Si is the projectile (see Figure 8*d*). Calculations treating ^{28}Si as inert, thus only accounting for the scaling effects due to the increased $Z_1 Z_2$, show the expected increased splitting of the two-peaked distribution (*solid line* in Figure 8*d*) as compared with that for $^{16}\mathrm{O} + {}^{144}\mathrm{Sm}$. This difference between the calculated and the experimental excitation function and barrier distribution indicates that couplings to excited states in ^{28}Si cannot be ignored. Realistic calculations were performed, including couplings to all orders and exact treatment of excitation energies, but retaining the isocentrifugal approximation. Still, the inclusion of the rotational states in ^{28}Si as well as the 2^+ and 3^- states in ^{144}Sm has not yet been able to reproduce all features of the data. Thus, while the barrier distribution gives us an additional handle on the type of couplings, they and the precise excitation functions place stringent demands on theoretical calculations to provide a complete picture of the reaction mechanism; an arbitrary inclusion of couplings or increased coupling strengths are thus no longer satisfactory as "explanations" of the observed sub-barrier fusion enhancement.

7.2 *The* ^{58}Ni + ^{60}Ni *Reaction*

The experimental study of ^{58}Ni + ^{60}Ni fusion (34) revealed for the first time the existence of a barrier distribution with several well-defined peaks that could only be explained by multiphonon couplings. The barrier distribution obtained by Stefanini et al (34), shown in the four panels of Figure 9, possesses two main peaks and a smaller structure at low energies. The *solid curve* in each panel shows the distribution resulting from coupled-channels calculations including an increasing number of vibrational phonon couplings. Because the single 2^+ quadrupole phonon has roughly the same energy in both nuclei, the coupling to these channels was replaced by a single effective channel (54) in the calculations (see also Section 4.2). The calculations employed the isocentrifugal approximation but took full account of the finite excitation energies and of Coulomb excitation.

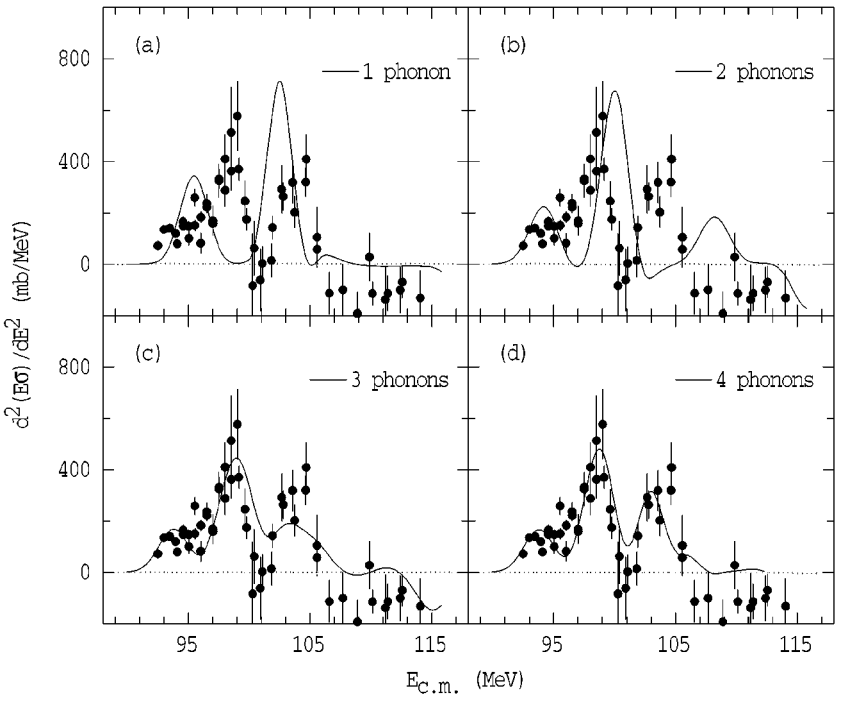


Figure 9 The fusion barrier distribution of ^{58}Ni + ^{60}Ni, compared with the results of coupled-channels calculations in the one-, two-, three-, and four-phonon channel spaces. Adapted from Ref. 34.

The calculations with one and two phonons (Figure 9*a, b*) are inadequate to reproduce the experimental evidence. Better agreement is found with three phonons, and further improvement results from adding the four-phonon channel, i.e. the mutual double-excitation of the quadrupole-phonon states. The full calculation gives a very good account of the measured excitation function down to the lowest measured cross sections of $\sigma \approx 50$ μb. It is interesting to note that the experimental data were reproduced without invoking coupling to the elastic pair-transfer channel, investigation of which was one of the original reasons for the measurement. Stefanini et al (34) justified this by pointing out the similarity in the experimental distributions for the ^{58}Ni + ^{60}Ni and ^{58}Ni + ^{58}Ni reactions. It was argued that the transfer couplings, which are different in these two systems, must have a small effect because the two measured distributions are so similar. In the light of this conclusion and the knowledge of the collective couplings acquired from the ^{58}Ni + ^{60}Ni barrier distribution, it would be interesting to see whether the data for the triad of reactions ^{58}Ni + ^{58}Ni, ^{58}Ni + ^{64}Ni, and ^{64}Ni + ^{64}Ni can now be explained consistently.

8. COUPLINGS TO TRANSFER CHANNELS

Earlier experimental studies of several reactions, for example ^{58}Ni + ^{64}Ni (19) and ^{32,36}S + ^{58,64}Ni (85), indicated the need to include couplings to single-, pair-, and multinucleon transfer in order to explain the observed fusion excitation functions. The importance of transfer in increasing the fusion probabilities was established, but quantitative understanding of the coupling strengths remained elusive due to the need to rely on complex theoretical calculations. Since the only observable was the cross section enhancement, isolating the effects of transfer couplings in the presence of couplings to inelastic channels (which are always present) required an accurate quantitative estimate of the enhancement due to inelastic channels alone. This was not always possible, as the ^{58,64}Ni + ^{58,64}Ni case demonstrates, because of the large number of channels involved and the uncertainties in calculating coupling strengths. The recent success in elucidating the relevant inelastic couplings that resulted from the measurement of the barrier distribution raises the following general questions: (*a*) Under which circumstances are the effects of transfer couplings apparent in barrier distributions? (*b*) Can the barrier distributions always be used to obtain a realistic estimate of the contributions from the inelastic channels, and hence an improved quantitative understanding of the transfer channels?

Most of the experiments performed to address these questions studied systems with positive *Q*-value neutron transfer reactions because they would give rise to a barrier lower in energy than the collective couplings. Since the collective couplings produce a higher-energy secondary peak, they were not expected

to interfere significantly with the conclusions. Neutron transfer is expected to be more probable and is easier to handle theoretically than proton transfer. The comparison of the measured (57) barrier distribution for the $^{17}O + {}^{144}Sm$ reaction with that for $^{16}O + {}^{144}Sm$ demonstrated unambiguously that coupling to the 1n stripping channel with a positive *Q*-value produced a barrier lower than the lowest barrier for $^{16}O + {}^{144}Sm$. The experiment showed that, although couplings to positive *Q*-value transfer channels are most evident in the enhancement of the cross sections at the lowest energies, the effects are also clearly apparent in the barrier distribution. The advantage of the latter is in constraining inelastic coupling strengths, thereby giving quantitative guidance for the transfer strengths. Many reactions have been studied since this early measurement, including $^{40}Ca + {}^{46,48,50}Ti$ (86), $^{32,36}S + {}^{110}Pd$ (87), $^{40}Ca + {}^{116,124}Sn$ (88), and $^{40}Ca + {}^{90,96}Zr$ (52, 53), and barrier distributions have been extracted to investigate the effects of two- or more neutron transfer channels with positive *Q* values. The $^{32,36}S + {}^{110}Pd$ and $^{40}Ca + {}^{90,96}Zr$ measurements are discussed, since they show typical effects of two-neutron and possibly multi-neutron transfer on fusion cross sections and barrier distributions.

8.1 *The* $^{32,36}S + {}^{110}Pd$ *Reactions*

The fusion of $^{32,36}S + {}^{110}Pd$ was studied at Legnaro (87) to investigate the couplings to the two-neutron transfer channel. All one- and two-nucleon transfer *Q*-values are negative in the case of $^{36}S + {}^{110}Pd$, whereas the two-neutron pickup channel has $Q_{gs} = +5.1$ MeV in the $^{32}S + {}^{110}Pd$ reaction. The common target, ^{110}Pd, is a vibrational nucleus with the lowest 2^+ state at 374 keV and $B(E2, 2^+ \rightarrow 0^+_{gs}) = 58$ W.u.; two- and possibly three-phonon states are known.

Figure 10 compares the measured excitation functions and the extracted barrier distributions on a reduced-energy scale. The barrier distributions for both systems have a very similar structure, with well-separated peaks, indicating the dominant effect of ^{110}Pd vibrational couplings. But barrier strength at low energies ($E/V_b \simeq 0.92$) is observed only in $^{32}S + {}^{110}Pd$, where it is qualitatively expected from strong positive *Q*-value transfer couplings. We note that this relatively small change in the barrier distribution causes the cross section for $^{32}S + {}^{110}Pd$ to be two orders of magnitude larger at low energies than it is for $^{36}S + {}^{110}Pd$.

8.2 *The* $^{40}Ca + {}^{90,96}Zr$ *Reactions*

Stelson et al (24) proposed that sequential transfer of several neutrons between the reactants may initiate fusion at large internuclear separations and will lead to flat barrier distributions. Timmers et al (52, 53) chose the $^{40}Ca + {}^{90,96}Zr$ systems to investigate such effects. Since the projectile ^{40}Ca is a magic nucleus, its influence on fusion was expected to be small (see also Section 9), and both

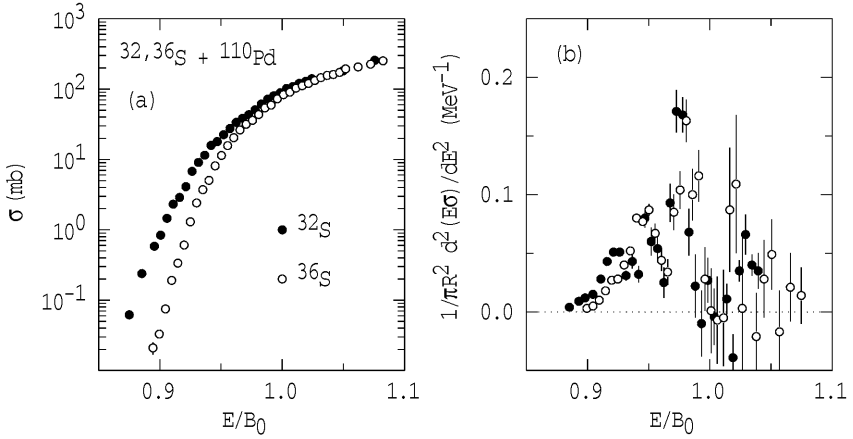


Figure 10 The excitation functions and the barrier distributions of $^{32,36}S + {}^{110}Pd$ on a reduced energy scale. The low-energy barrier strength is present only in the case of $^{32}S + {}^{110}Pd$, which has a positive *Q*-value for the two-neutron pickup channel. Adapted from Ref. 87.

$^{90,96}Zr$ nuclei have similar shell structures with moderately collective 2^+ and 3^- states. Because the main difference between the two reactions is the neutron transfer *Q*-values, their effects should be clear.

Figure 11*a* shows the two measured excitation functions on a reduced-energy scale. A very large relative enhancement is observed for $^{40}Ca + {}^{96}Zr$ at low energies. The barrier distribution plotted in Figure 11*c* shows a flat and rather structureless distribution for ^{96}Zr, extending to very low energies. It is qualitatively different from the distribution of ^{90}Zr (see Figure 11*b*), which shows well-defined peaks. The barrier structure and the fusion excitation function for the reaction $^{40}Ca + {}^{90}Zr$ could be explained in terms of the coupling to single- and double-phonon excitations of ^{90}Zr (solid lines in Figure 11*a, b*), but these couplings could not explain the fusion of $^{40}Ca + {}^{96}Zr$ (dashed curves in Figure 11*a, c*). Since no other inelastic channel can modify the situation, the differences were attributed to strong coupling to few- and possibly multineutron transfer channels in $^{40}Ca + {}^{96}Zr$, where single-, pair-, and multineutron transfer *Q*-values are very positive. The corresponding *Q*-values in $^{40}Ca + {}^{90}Zr$ are all negative. The neutrons in ^{90}Zr are essentially frozen in the full $N = 50$ shell, but it is favorable to transfer the "valence" neutrons in ^{96}Zr to ^{40}Ca, approaching the $N = 28$ shell. Calculations (53) indicate that sequential neutron transfer, rather than single-step neutron transfer, is required to explain the observed distributions. In the absence of a full coupled-channels calculation, it is perhaps premature to take the results as conclusive evidence of sequential

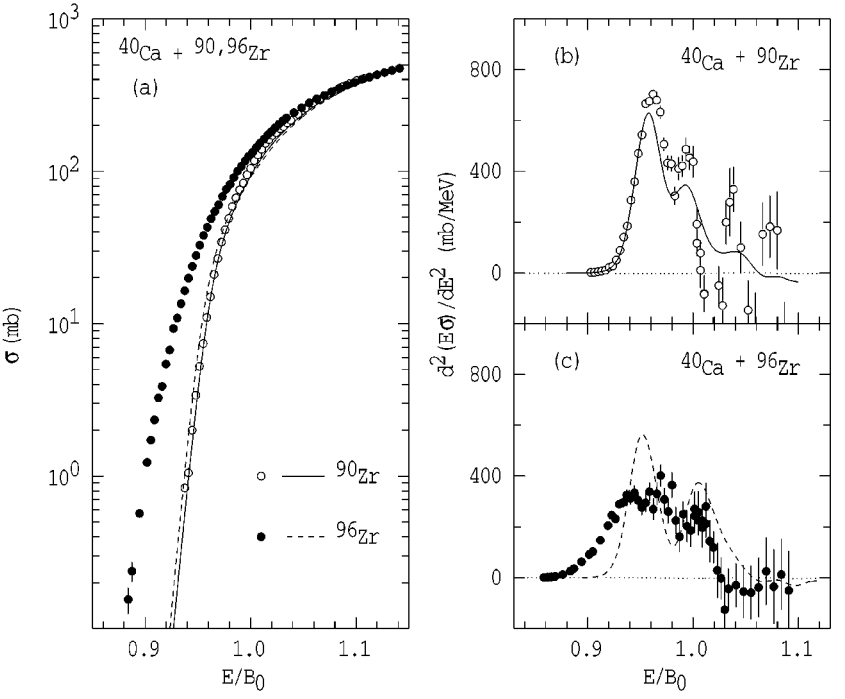


Figure 11 The fusion excitation functions and barrier distributions for ^{40}Ca + ^{90,96}Zr reactions. The coupled-channels calculations for ^{90}Zr (*solid*) and ^{96}Zr (*dashed*) including up to four-phonon excitations of the Zr nucleus are shown. Adapted from Ref. 52.

neutron transfer. However, the results gain support from the recent measurements (89) of multineutron transfer reactions in both systems at an energy slightly above the Coulomb barrier, where much larger transfer probabilities have been found for the heavier target.

8.3 *Single-Particle and Collective Aspects of Transfer*

For reactions where couplings to positive *Q*-value one- or two-neutron transfers are important, the experimental evidence shows that while they are mainly responsible for large cross section enhancement (or change of slope) at the lowest energies, these transfers do not change the barrier distributions significantly. In the presence of collective couplings (e.g. rotations or vibrations), their effect is to introduce a barrier, usually with a small weight, which is lower than the main barrier. This gives rise to substantial cross section enhancement at the lowest energies. The small effect of such transfers on the overall barrier

distribution seems to imply that negative Q-value transfer reactions may also not significantly alter the distributions above the main barrier, and hence the conclusions regarding the couplings to inelastic channels discussed in the previous sections should not be materially affected. Similar conclusions regarding the small strength of transfer compared with collective excitations were also drawn by others (90–92) in fitting fusion excitation functions. This scenario, however, changes when multinucleon transfers are involved. Their effect on cross sections and barrier distributions becomes very significant at energies below and close to the barrier, as evident from the comparison of the data for the two reactions ^{40}Ca + ^{90,96}Zr.

There have been attempts to correlate transfer cross sections and fusion enhancements; indeed, much effort has been invested in measuring the transfer cross sections, and in some cases [e.g. (93–95)] extracting an effective coupling strength, which could then be used in calculating the fusion cross sections. It is not clear how strong the correlation is, as the coupling strengths exactly at the barrier influence fusion, whereas the transfer cross sections result mainly from more peripheral interactions. However, if the coupling strengths and form factors are known well (or can be calculated), a complete coupled-channels calculation should be able to predict both fusion and transfer reactions (and inelastic scattering) in a consistent manner. Due to the paucity of complete data sets, and the complexity of the calculations, such analyses are scarce (62). The recent availability of precise fusion data, as well as transfer yields, now makes it appropriate simultaneously to analyze fusion and few-nucleon transfer data in a realistic fashion. This effort would provide a stepping stone for the description of correlated or sequential transfers that can be thought of as a precursor of neck formation, a phenomenon of interest in the formation of heavy elements.

9. COUPLINGS TO HIGH-LYING STATES

As seen in the previous sections, a successful description of the fusion cross sections for reactions involving ^{16}O did not require coupling to excited states in ^{16}O, and the same should hold for reactions with ^{12}C. This is despite the fact that these nuclei have very collective high-lying quadrupole and octupole states. In the eigenchannel approximation, coupling to such states gives rise to the usual barrier splitting (see Section 3.2), but the lower barrier has a weight close to unity, while that of the higher barrier is negligible; the lowering of the main barrier compared with the uncoupled barrier can be considered a "polarization" contribution (9) to the potential.

It was therefore disconcerting to find that, for ^{16}O + ^{144}Sm, the good theoretical representation of the experimental fusion barrier distribution obtained

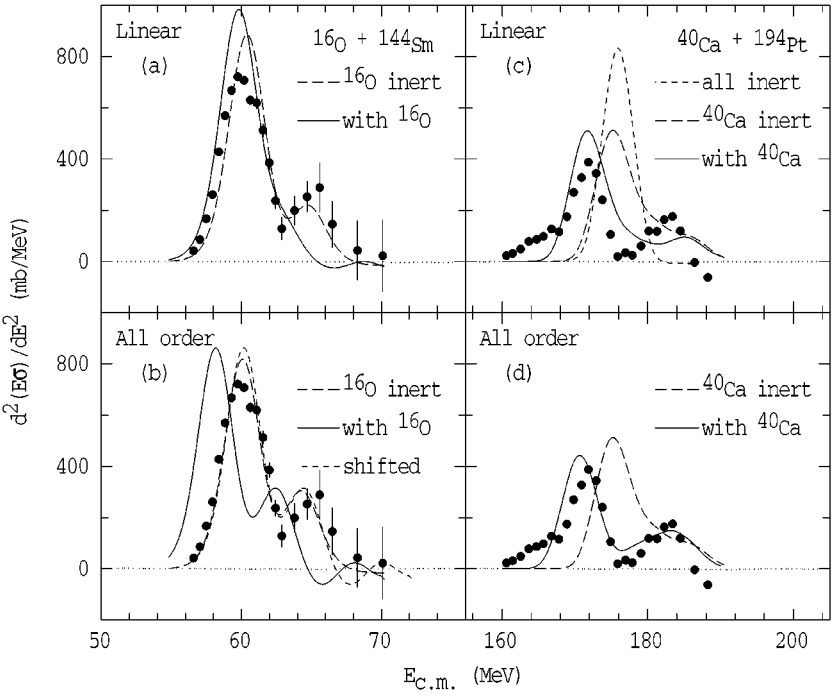


Figure 12 The barrier distributions for the $^{16}O + ^{144}Sm$ and $^{40}Ca + ^{194}Pt$ system compared using realistic coupled-channels calculations with the linear coupling approximation (*upper panels*) and all-order coupling (*lower panels*). Adapted from Ref. 96.

by ignoring couplings to the 3^- state in ^{16}O (as shown by the *dashed line* in Figure 12*a*) is destroyed when the projectile excitation is included (*solid line*). This was particularly curious because barrier distributions for $^{40}Ca + ^{192}Os$, ^{194}Pt reactions, which involve another closed-shell nucleus (^{40}Ca), showed a characteristic structure with a high energy peak, indicated by the vertical arrows in Figure 7, which could only be explained by including coupling to the octupole state of ^{40}Ca. It can be argued that the octupole state in ^{40}Ca has more effect because the energy is lower (3.7 MeV) than that of the octupole state in ^{16}O (6.1 MeV) and the coupling strength is larger due to the larger $Z_1 Z_2$. Nevertheless, the problem of reconciling the discrepancy for $^{16}O + ^{144}Sm$ remained.

Hagino et al (96) pointed out that the observed discrepancy is due to the inadequacy of the linear coupling approximation, used in the above calculations, for calculating couplings to states with large collective E2 or E3 strengths. Figure 12*b* shows the results of their coupled-channels calculations (full curve),

including couplings to ^{16}O states to all orders as well as to the 2^+ and 3^- states of ^{144}Sm. The calculation shows a shift in energy but reestablishes the double-peaked structure seen in the experimental data, which was absent in the equivalent linear coupling calculations. This shift is consistent with the general conclusion that the main effect of coupling to an inelastic channel with an excitation energy larger than the curvature of the bare fusion barrier is the introduction of a static potential shift (97). The shape of the barrier distribution does not change unless the coupling is very strong and the coupling form factor itself has a strong radial dependence. These results can be understood intuitively if we consider that the high excitation energy of the state in the projectile corresponds to a low moment of inertia; thus the projectile responds so quickly to the forces on it that it always arrives at the barrier in the lowest-energy configuration, and thus only shifts the barrier without changing its shape. The effects of these high-lying states can be included in the "bare" potential in the coupled-channels calculations. Thus, for the ^{16}O + ^{144}Sm calculation, where the potential parameters were obtained from a single-barrier fit to the high-energy data, the effects of the octupole vibration of ^{16}O are implicitly included. The explicit inclusion of this coupling then leads to double counting, which manifests itself as an additional shift in the barrier (or barrier distribution) to lower energy.

In the case of the ^{40}Ca + ^{192}Os, ^{194}Pt reactions, calculations including all-order couplings to the rotational band in the target nuclei, but ignoring the excitation of ^{40}Ca, give results similar to those expected for a classically deformed nucleus. However, these are inconsistent (77, 96) with the experimental results, as shown by the long dashed curves in the right panels of Figure 12 for the case of ^{40}Ca + ^{194}Pt. When the calculations include the octupole excitation of ^{40}Ca in the linear coupling approximation, the shape of the barrier distribution is significantly changed, and a higher energy peak is introduced (Figure 12*c*). This calculation reproduces qualitative features of the data but does not achieve quantitative agreement. The calculations by Hagino et al (96), which treat the couplings to the 3^- state in ^{40}Ca to all orders (Figure 12*d*), produce a higher energy peak in quantitative agreement with the experimental observation. It is interesting to note that for this reaction, unlike the case of ^{16}O + ^{144}Sm, the qualitative features of linear and all-order coupling calculations are similar, as a result of the combination of the smaller excitation energy and β_3 of the 3^- state in ^{40}Ca. The theoretical calculations still significantly underestimate the barrier strength at low energies because these calculations do not take into account couplings to transfer channels, which are expected to be present.

The calculations with full-order coupling thus show that, while the dominant effect of the excitation of the ^{16}O octupole state at 6.1 MeV is simply to renormalize the static potential, the coupling to the 3^- state in ^{40}Ca, situated

at a relatively lower energy of 3.7 MeV, introduces well-defined peaks in the barrier distribution for ^{40}Ca + ^{192}Os, ^{194}Pt. For reactions of ^{40}Ca with lighter targets such as ^{40}Ca (32) and ^{90,96}Zr (52), where the coupling strengths are small due to the low Z_1Z_2, coupling to the 3^- state of ^{40}Ca only renormalizes the static potential without changing the shape of the barrier distribution. The above calculations thus resolved the apparently conflicting conclusions regarding the influence of coupling to states at high excitation energy. More importantly, these calculations suggest a natural limit to the energy of states which need to be considered explicitly in coupled-channels calculations. The myriad of weak high-energy excitations contribute only to a potential renormalization without affecting the shape of the barrier distribution at all. The effects of these excitations can then be included in the bare potential in coupled-channels calculations.

10. BARRIER DISTRIBUTIONS FROM SCATTERING DATA

The coupled-channels method describes heavy-ion reactions in a unified way, predicting cross sections for all channels specifically included in the calculation. One might, therefore, expect the distribution of barriers to be reflected in all channels.

It was recently shown (98) that a representation of barrier distributions might be extracted from quasi-elastic scattering differential cross sections $d\sigma_{qel}/d\Omega$ measured at backward angles (comprising all reaction products that do not pass over the fusion barrier, and thus including inelastic and transfer channels). This requires a single differentiation of the ratio to the Rutherford cross section ($d\sigma_R/d\Omega$) at the same angle:

$$D_{qel}(E) = -\frac{d}{dE}\left(\frac{d\sigma_{qel}}{d\sigma_R}\right). \qquad 42.$$

The major advantage of this procedure is the possibility of smaller experimental uncertainties, particularly at energies above the barrier, since $d\sigma_{qel}/d\sigma_R$ is decreasing rapidly with increasing energy here, similar to the trend in fusion cross sections as the energy decreases below the barrier.

Measurements for ^{16}O + ^{92}Zr, ^{144,154}Sm, ^{186}W (98) showed that, although the general features of the barrier distributions are retained in the representations obtained from quasi-elastic scattering, some features are lost. The barrier distribution determined from the fusion data (D_{exp}) for ^{16}O + ^{144}Sm (see Figure 8) exhibited a double-peaked structure, whereas D_{qel}, as seen in Figure 13*a*, showed a smooth fall above the peak in the distribution. Exact coupled-channels calculations including only inelastic excitations of ^{144}Sm (*solid curve*) indeed showed the two peaks expected; however, it was noted

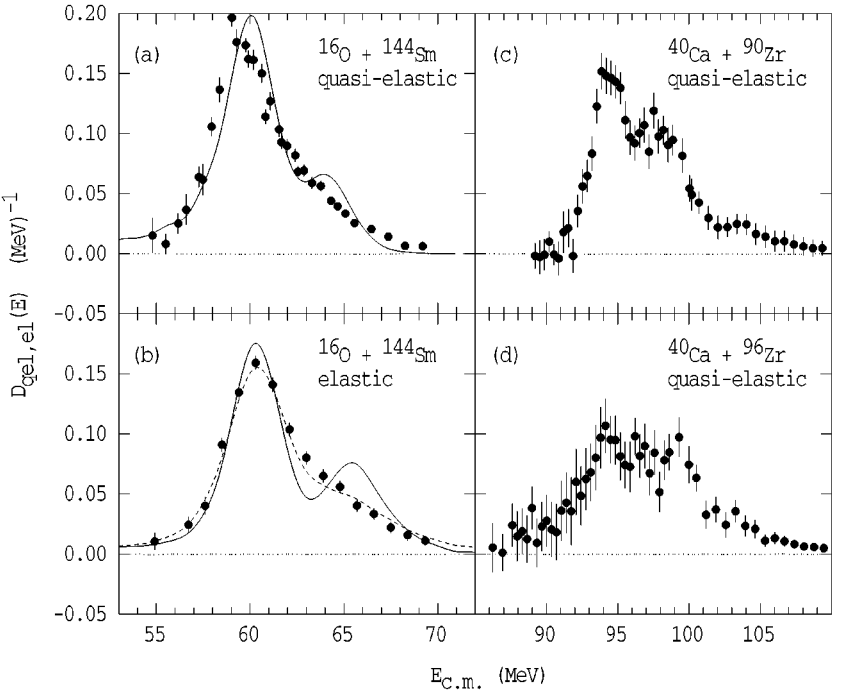


Figure 13 Experimental barrier distributions obtained from quasi-elastic scattering data for the reactions indicated. Data are from Refs. 53, 98.

that the measured quasi-elastic yield comprised largely transfer products at the higher energies. It was concluded that "de-phasing" due to kinetic energy and angular momentum differences between the initial and final channels was responsible for the loss of resolution.

A later simultaneous measurement of fusion and backward–angle quasi–elastic scattering for the reaction $^{40}Ca + {}^{90,96}Zr$ (53) showed closer correspondence between the two representations of the barrier distribution. In particular, the structures near and below the average barrier present in D_{exp}, seen in Figure 11(*b*) and (*c*), appear quite convincingly in the D_{qel} shown in Figure 13(*c*) and (*d*). The fact that this structure is not above the average barrier, as it is for $^{16}O + {}^{144}Sm$, may contribute to the agreement, since the quasi–elastic yield is not so dominated by transfer reactions.

To eliminate the energy and angular momentum differences suspected as the cause of loss of resolution in D_{qel}, another representation of the barrier distribution, determined from pure elastic scattering, was proposed (99), where

it was shown that within the eigenchannel approximation,

$$D_{el}(E) = -\frac{d}{dE}\left(\frac{d\sigma_{el}}{d\sigma_R}\right)^{1/2}. \qquad 43.$$

However, the measured elastic scattering excitation functions for $^{16}O + ^{144}Sm$ showed that D_{el} is again smooth at the higher energies (see Figure 13*b*). This was explained by invoking the presence of weakly coupled channels, which, in the language of the optical model, constitute a surface imaginary potential. Calculations using the coupled-channels code ECIS (100), including inelastic excitations alone, predicted a double-peaked distribution (full curve), but inclusion of a surface imaginary potential resulted in a good reproduction of the measured D_{el} for $^{16}O + ^{144}Sm$, as shown by the dashed line.

In contrast with fusion, quasi-elastic and elastic scattering is phase sensitive. This makes it difficult to interpret the barrier distributions measured in these channels. It is, however, significant that the data show a strong relationship between D_{qel}, D_{el}, and D_{exp} as expected in the eigenchannel picture. It would be interesting to compare these representations for systems having larger Z_1Z_2 with pronounced barrier structure spread over a larger energy range.

11. EXPLOITING FUSION BARRIER DISTRIBUTIONS

The existence of a distribution of fusion barriers has consequences for many other channels. Indeed, the previous section has demonstrated that it might be more appropriate to refer to these as reaction barriers, since the same set of barriers determines elastic and quasi-elastic scattering, etc. There are, however, also important consequences for the fusion channel itself. Most notably, the spin distribution of the compound nucleus is strongly affected by the presence of several barriers. A review in 1992 of angular momentum distributions of the compound nucleus (101) considered many aspects of this relationship.

11.1 *Angular Momentum Distributions*

Various models have shown that the fusion transmission coefficient for each ℓ-wave (or the angular momentum distribution) can be mapped from the barrier distribution (102). Equivalently, a model that reproduces the barrier distribution and fusion excitation function is expected to predict accurately the angular momentum distribution of the fused nuclei. Conversely, the barrier distribution can be mapped from the spin distribution measured, in principle, at a single beam energy with sufficient precision. Experiments have been performed with this goal (103), measuring gamma-ray multiplicity distributions. Very high

statistics must be obtained, and reactions and beam energies must be chosen with care, but recent results are encouraging.

A reliable knowledge of angular momentum distributions is useful to explain and predict cross sections for populating super-deformed bands for nuclear spectroscopy studies (104). Intuitively, the lowest-energy fusion barrier should generally correspond to an elongated configuration of the two nuclei. It will be interesting to investigate whether the different configurations corresponding to the different barriers influence the shape of the compound nucleus (105). Does the strongly deformed configuration corresponding to the lowest entrance-channel barrier preferentially lead to highly deformed (super- or hyperdeformed) gamma-ray emitting states? A further question of interest is whether we can take advantage of low energy barriers to produce exotic evaporation residues. Some of these points are under preliminary investigation in the ^{58}Ni + ^{60}Ni system discussed above (S Courtin, private communication).

A good knowledge of the spin distribution is also necessary for the accurate interpretation of measured fission probabilities in heavy-ion reactions (106), since the fission probability increases rapidly with angular momentum. Below we describe in detail, for two related examples, applications of the improved quantitative understanding of fusion angular momentum distributions that resulted from fusion barrier distribution measurements. These were unsolved problems at the time of the 1992 review of angular momentum distributions (101).

11.2 *Fusion-Fission Angular Anisotropies*

Fission angular distributions depend strongly on angular momentum and have been used both to study fission, where the angular momentum distribution is thought to be known, and to study fusion, where it is not. The fission angular distribution can be characterized by the anisotropy A, defined as the ratio of yields at 180° or 0° to that at 90° to the beam axis. The anisotropy is approximately given by the expression, based on the transition state model (TSM) (107),

$$A = 1 + \frac{\langle \ell^2 \rangle \hbar^2}{4 I_{eff} T}. \qquad 44.$$

Here I_{eff} is the effective moment of inertia at the saddle point and T is the temperature, while $\langle \ell^2 \rangle$ is closely related to the mean square angular momentum following fusion. The equation shows that A is very sensitive to $\langle \ell^2 \rangle$, as well as to properties of the fission process. A long-standing puzzle for the reaction ^{16}O + ^{208}Pb, where the TSM was expected to be applicable, was the extraction using this equation of large values of $\langle \ell^2 \rangle$ at sub-barrier energies (108), inconsistent with fusion models. Morton et al (72) performed a more precise

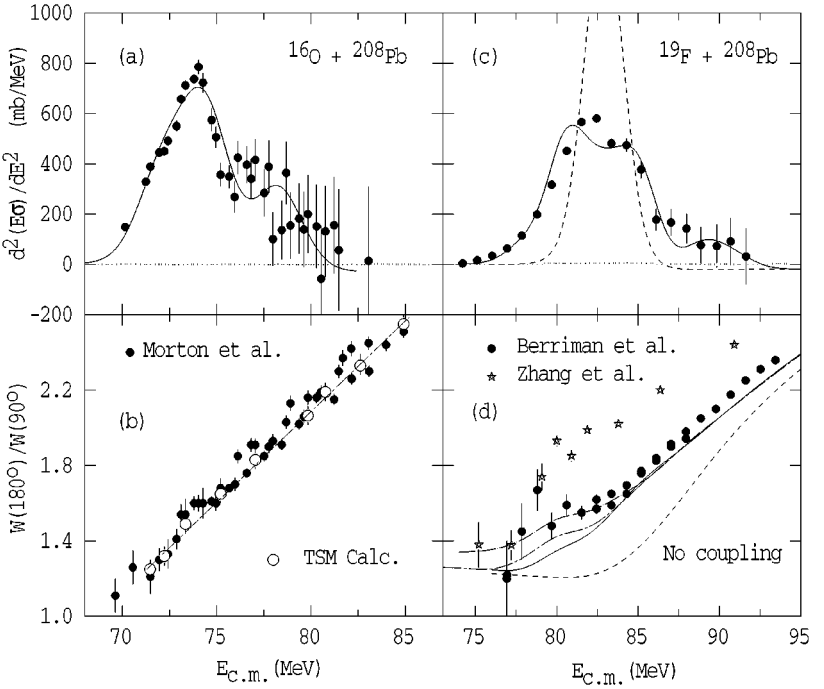


Figure 14 Panels (*a*) and (*c*): The measured fusion barrier distributions, and empirical calculations (*solid lines*) that reproduce them, for the ^{16}O, $^{19}F + ^{208}Pb$ reactions. Panels (*b*) and (*d*): The fission anisotropies. Calculated anisotropies based on the transition state model (TSM) are shown; those for $^{19}F + ^{208}Pb$ illustrate the sensitivity to different coupling scenarios. Data are taken from Refs. 72, 110.

remeasurement of both the anisotropies and fusion cross sections; they used an empirical calculation reproducing the fusion cross sections and the barrier distribution (see Figure 14*a*) to predict the fusion angular momentum distributions. These then were used as input in a detailed Monte Carlo calculation of the fission anisotropies. Due to improvements in both the data and the calculations, agreement was obtained (see Figure 14*b*), indicating that the fusion and TSM models are both valid for this reaction (72).

Because anomalously large anisotropies had also been reported at near-barrier energies for $^{19}F + ^{208}Pb$, this reaction was also remeasured (109, 110) to investigate whether quasi-fission could be responsible, as proposed for projectiles with $A \geq 24$ incident on ^{208}Pb (111). The experimental results are shown in the right panels of Figure 14. The measured anisotropies in the fusion barrier region were lower than those previously reported, and the calculations gave an

excellent reproduction of the new data. The anisotropies at sub-barrier energies are sensitive to details of the fusion calculation, e.g. to the fusion barrier curvature, and to the energy of the lowest barrier, the three calculations showing that the variation in predicted anisotropy is measurable.

With smaller experimental uncertainties, anisotropies at far sub-barrier energies could become a valuable tool to clarify or confirm the understanding of the fusion process resulting from fusion cross section data alone. At near- and above-barrier energies, the information on the fusion angular momentum distribution that can be extracted from the measured barrier distribution may allow the anisotropies to give information on fission dynamics (109), which was previously inaccessible due to uncertainty in the angular momentum distribution.

11.3 *Fusion-Fission and Quasi-Fission Competition*

Measured angular distributions for fission following bombardment of heavy targets ($Z \geq 82$) with heavy projectiles ($Z \geq 12$) exhibit larger anisotropies (111) than predicted by the TSM. This has been attributed to quasi-fission (63, 111), but, in a manner similar to fusion-fission, decreasing bombarding energies resulted in a decrease of the anisotropies (111).

For reactions with actinide targets at near-barrier energies, a different and puzzling behavior was persistently observed. The anisotropies increased (108, 112–116) to values much larger than TSM predictions as the beam energy dropped through the fusion barrier region. A method developed (79) to isolate fission following capture in the fusion potential pocket from fission of target-like nuclei, using the deduced velocity vector of the fissioning nucleus, resulted in more reliable data and helped to clarify this anomaly. Figure 15 shows fusion barrier distributions and anisotropies for the ^{12}C, $^{16}O + ^{238}U$ reactions (79, 117).

The shapes of the fusion barrier distributions are very similar in these two reactions, as might be expected. Simplified coupled-channels calculations (56) using the tabulated deformation parameters for ^{238}U reproduced them reasonably well. This established experimentally that the lowest fusion barriers correspond to collisions only with the elongated tips of the target nuclei. Reactions at sub-barrier energies thus appear to be with an aligned target.

The fission anisotropies (Figure 15*b, d*) show the same qualitative features determined in other measurements (108, 112, 113, 115, 116, 118), although quantitative differences occur. The expected energy dependence of the anisotropy based on TSM calculations is indicated by the dot-dashed curves, and no reasonable variation of parameters could reproduce the data. The good agreement of data and TSM-based calculations obtained for the reactions ^{16}O, $^{19}F + ^{208}Pb$ suggests that neither failure of the fusion models nor breakdown of the

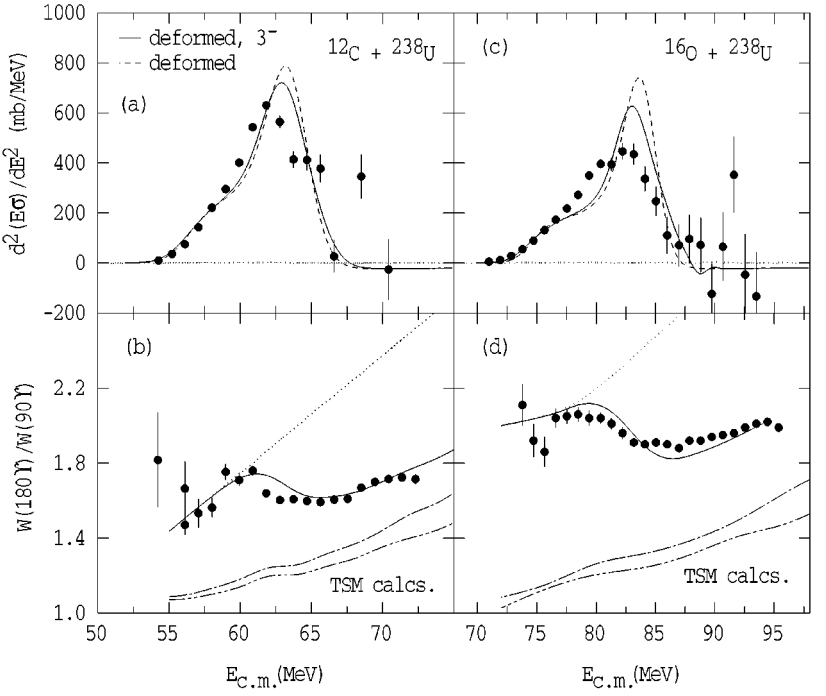


Figure 15 Panels (*a*) and (*c*): Barrier distributions for the ^{12}C, ^{16}O + ^{238}U reactions. Calculations for the measured β_2 and β_4 with and without coupling to the 3^- state in ^{238}U are shown. Panels (*b*) and (*d*): The fission anisotropies. Calculated anisotropies based on different assumptions about the fission dynamics are indicated (TSM); the *dotted line* gives the assumed anisotropy for quasi-fission. The model described in the text gives the *solid curves*, which agree well with the data. Adapted from Refs. 79, 118.

TSM itself is responsible for this feature. It was noted (114) that the deviations of the data from the prediction showed a striking correlation with the fusion barrier distribution, suggesting a relationship between the anisotropy and the relative orientations of the colliding nuclei.

One explanation for the anomalous experimental results was based on the suggestion that the physical configuration (orientation) of the target nucleus and projectile at the fusion barrier influences the subsequent reaction dynamics (79, 114). This explanation assumed that collisions with the tips of the target nuclei result in quasi-fission only. The data at the lowest energies then defined the angular anisotropies for quasi-fission, with an assumed energy dependence shown by the *dotted lines* in Figure 15. A transition from quasi-fission to fusion-fission was assumed to occur at a critical angle between the beam axis

and the target nucleus symmetry axis (equivalent to a critical radius). Taking weighted averages of the anisotropies assumed for quasi-fission and those calculated for fusion-fission, for a critical angle of 35°, results in the anisotropies shown by the *solid lines*. These follow the trend of the data very well, supporting the hypothesis that collisions with the tips of the deformed target nuclei tend to result in quasi-fission while collisions with the sides result rather in formation of a compound nucleus (almost always followed, however, by fission decay).

An alternative explanation (119), also based on the relationship between the beam energy and the orientations of the deformed target that will result in capture, assumes that fission occurs along the deformation axis of the target nucleus. This model postulates a memory of the initial *K*-value at the time of capture, where *K* is the projection of the total angular momentum on the scission axis. At the lowest energies, *K* is thus zero, and so the fission anisotropy is large; at energies above the barrier region, all possible values of *K* occur. Data for $^{12}C + ^{235,238}U$ support this hypothesis, but this memory effect also implies that a true compound nucleus is not formed. It will be interesting to see whether a combination of these two models can describe both the magnitude and the energy dependence of the anisotropies. It seems likely that a full dynamical calculation of both the shape and the *K* evolution is necessary, a challenge to theoretical modeling (and computational power), and probably a severe test of any transport model.

For quasi-fission at center-of-mass angles other than 90°, there should be some memory of the projectile mass and direction, resulting in an asymmetry in the mass distribution. Measured fission mass distributions for the $^{16}O + ^{238}U$ reaction at a mean center-of-mass angle of 130° displayed a slight but significant mass asymmetry. Although this was good evidence for the presence of quasi-fission, the energy dependence (Figure 16*a*) is more convincing, since it also could be described by the same critical angle model that described the angular anisotropies. A measurement for the reaction $^{32}S + ^{232}Th$ (120) investigated the role of deformation of the target in a reaction where quasi-fission is dominant. The data (see Figure 16*b*) show a strong variation of the mass centroid with beam energy through the fusion barrier region. This suggests that the orientation of the actinide nucleus also has a strong effect on the characteristics of the quasi-fission process. The energy dependence of the angular anisotropies further supports this picture.

These data indicate that the configuration (particularly elongation) at the fusion barrier has a strong influence on the subsequent dynamics. They suggest that forming spherical compound nuclei at low excitation energy in sub-barrier collisions of deformed nuclei is not favorable, and they support the theoretical suggestion (121) that collisions of deformed nuclei at energies above the

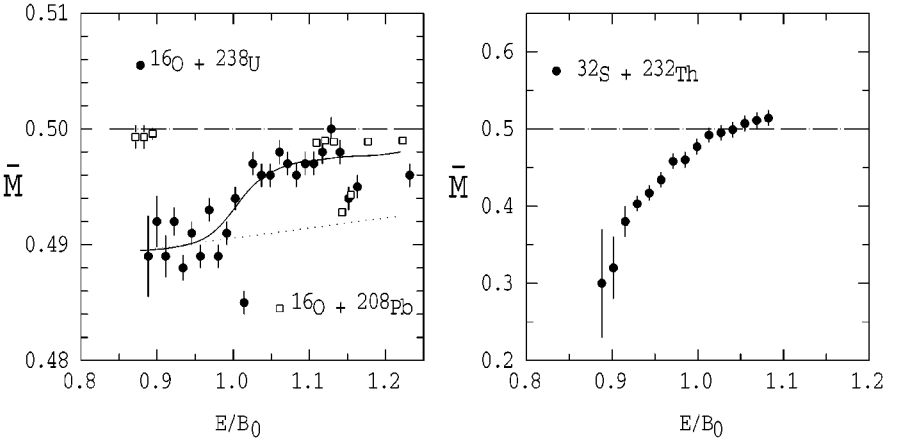


Figure 16 The FMT fission mass centroids for $^{16}O + ^{238}U$ and $^{32}S + ^{232}Th$ as a function of the ratio of the beam energy to the single-barrier energy. The expectations based on the statistical decay model are shown by the *dot-dashed line*. The calculation (*full curve*) that reproduces the energy dependence of the $^{16}O + ^{238}U$ data is described in the text. Data are from Refs. 79, 120.

mean fusion barrier may be more promising. However, the larger excitation energy that is inevitably introduced allows more opportunities for the compound nucleus to decay by fission, suppressing the survival of evaporation residues; evaporation-residue excitation functions for very heavy nuclei suggest that this disadvantage may well outweigh the entrance-channel effects described above.

12. BREAKDOWN OF ENTRANCE-CHANNEL MODELS

It has been noted that passage inside the fusion barrier is not a sufficient criterion to guarantee formation of a compound nucleus. Swiatecki (122) modeled the evolution of the nuclear shape after passage inside the barrier in an elegant and simple qualitative picture. This model of nuclear coalescence predicted a dramatic change in reaction mechanism for collisions with charge product $Z_1Z_2 \geq 1600$. Below this threshold, passage inside the fusion barrier (the first barrier to fusion) guarantees compound nucleus formation. Above this value, extra energy was found necessary to push the system to more compact configurations. In this model, the second barrier to be overcome on the route to compound nucleus formation is the conditional (fixed mass asymmetry) fission saddle point, which, for heavy systems and small entrance-channel mass asymmetries, is more compact than the fusion barrier configuration. The

energy dissipation in this model (taken to be one-body dissipation) necessitates an extra kinetic energy, called the "extra push," to pass inside the conditional saddle point. To form a compound nucleus at its equilibrium configuration, it is necessary to reach a configuration more compact than the unconditional (symmetric fission) saddle point, which is itself more compact than the conditional saddle point. Depending on the system and hence the potential energy surface, this may require even more kinetic energy, called the "extra-extra push." Those trajectories trapped initially inside the conditional saddle point, but outside the fission barrier, are expected to result in quasi-fission. The calculated extra- and extra-extra–push energies depended largely on the elongations of the fusion barrier and the conditional and unconditional saddle points, as well as on the strength and form of the frictional forces assumed to be operating. The underlying reason for the extra-push effect, however, is the large Coulomb repulsion felt by the heavy system in any configuration.

Experimental data for a range of reactions (9, 63) showed general agreement with the predictions, although the onset of the extra-extra push was smoother than predicted. These data and the concepts of the model indicated that true fusion cannot be described only in terms of the entrance channel for such reactions. Fusion probabilities were extracted from some of the measured evaporation-residue excitation functions (27, 123), and information on the width of the fusion barrier distributions. A correlation was found between the size of the shift of the barrier from the Bass model (124) predictions and the width of the barrier distributions, and these were correlated with the nuclear structure of the projectile and target.

Such a dependence on nuclear structure effects was not built into Swiatecki's model. Alternative entrance-channel models were developed that could describe the smooth onset of the extra push [the surface friction model (125)] or the correlation of barrier widths and extra-push energies [the dissipative diabatic dynamics model (126)]. These models, however, do not describe the quasi-fission mechanism or the effects seen in the fusion barrier distributions discussed in this review. It has been suggested (127) that evidence exists in the measured fusion barrier distribution for ^{58}Ni + ^{60}Ni for an entrance-channel mechanism that can contribute to the extra-push energy. Roughly speaking, the three extracted barriers in this system can be thought of as follows: (*a*) The lowest corresponds to induced prolate deformations of the two nuclei along their line of centers, thus lowering the barrier; (*b*) in the configuration corresponding to the central barrier the two nuclei remain more or less spherical; (*c*) the high barrier corresponds to aligned oblate deformations. All of these different configurations exist in the entrance-channel wavefunction, and the Coulomb barrier acts as a filter allowing the successive fusion of the different components as the energy increases. The effect of Coulomb excitation is a dramatic

reduction in the weight of the low-energy barrier. The energetically favorable oblate deformation in the repulsive Coulomb field is the least favorable one for the attractive nuclear force and is preferentially populated at large distances before the nuclear forces come into play. Calculations show that, as the charge product $Z_1 Z_2$ increases, this effect becomes larger, with a significant shift of the weight of the barriers into that with the highest energy. This produces a barrier centroid that lies above the usual Bass value, giving an extra-push effect.

It may be that a combination of the coupled-channels approach in the entrance phase of the reaction with a transport model describing the subsequent diffusion over the potential energy surface (128) would give the best description of the complex dynamics governing the compound nucleus formation process for reactions between heavy nuclei. The critical point in this marriage would be the boundary between the two very different models. The question is to what extent the fusion barrier energy can be correlated with the physical configuration (particularly elongation) of the system at the starting point of the thermal diffusion. For the case of deformed nuclei, the experimental results described in Section 11.3 suggest a strong correlation of the outcome of the reaction on the orientation of the deformed heavy nucleus, while the above discussion of vibrational systems implies that there may be a similar correlation here.

There is evidence that the dominant role played by the collective excitations characteristic of the individual colliding nuclei (rotation and vibrations), which is clear for lighter systems, may become subsumed in heavier systems by a new collective phenomenon, relevant only to the combined system, namely neck formation. This phenomenon seems to need further investigation. Measurements of the "capture barrier distributions" for such heavy systems, rather than the distributions for true fusion, may allow a better understanding of the role of neck formation in such reactions.

13. FUSION OF RADIOACTIVE NUCLEI

The new generation of radioactive beam facilities currently in development will permit measurements of heavy-ion fusion reactions with "exotic" nuclei. Ideally these reactions should be measured to the same level of precision as is currently possible for stable beams. However, these beams have a significantly lower intensity, and obtaining the necessary precision will be difficult. To date, few fusion experiments have been performed, and these have mainly involved rather light beams for which $Z_1 Z_2$ is relatively small. For such systems, the barrier distribution is narrow, making it difficult to extract information. However, the insights into fusion dynamics already gained in the study of barrier

distributions for heavier systems should now help in the analyses of the new systems, for which many new features may appear. Heavier exotic systems may possess neutron skins, and one way of measuring the extent of these skins would be to measure the fusion threshold. It is clear, however, that a good understanding of the effects of the relevant collective modes in the stable isotopes of the same element will be necessary in order to extract the desired radius information. The understanding of fusion of radioactive nuclei is necessary for the effective use of such reactions as a tool to produce new nuclei far from stability.

13.1 *Fusion of Weakly Bound Nuclei*

The presence of a neutron halo in light neutron-rich nuclei is expected to lead to a lowering of the mean fusion barrier due to the greatly increased radial extent of the nuclear matter distribution. The presence of a halo also increases the likelihood of coupling to excitation modes that are not significant in more stable nuclei. Coupled-channels calculations that include coupling to the soft dipole oscillation (129) show an enhancement of sub-barrier fusion cross sections. The small binding energy, however, means that such nuclei can easily be excited above the breakup threshold. This will lead to a suppression of the complete fusion yield, which will depend strongly on the radial dependence of the breakup probability.

Models exist that treat one or more of the aspects discussed above, but their predictions differ in detail. To compare models with experiment, both sub-barrier and above-barrier fusion cross sections must be available to allow investigation of both enhancement and suppression mechanisms.

Despite the limited beam intensities currently available, fusion excitation functions in the barrier region have been measured for $^{9,11}\mathrm{Be} + {}^{238}\mathrm{U}$ (130) and $^{9,10,11}\mathrm{Be} + {}^{209}\mathrm{Bi}$ (131). Each study has included the reaction with the stable $^{9}\mathrm{Be}$ for comparison with the radioactive isotopes. In the reactions on $^{238}\mathrm{U}$, fission of the target-like nucleus following peripheral reactions may confuse the picture, so the reaction with Bi should give a clearer picture. The experimental data for these reactions, however, show that the fusion excitation functions are similar for the $^{10,11}\mathrm{Be}$ reactions, while $^{9}\mathrm{Be}$ actually gives the lowest fusion yield, as shown in Figure 17*a*. It is not obvious whether the difference is due to differing fusion enhancement or to suppression. Experimental measurements of fusion cross sections at energies well above the fusion barrier (132, 133) for reactions on lighter targets show that fusion cross sections are suppressed for reactions with $^{9}\mathrm{Be}$; thus, it may be dangerous to use $^{9}\mathrm{Be}$ as a standard against which to judge the behavior of the radioactive isotopes.

To determine precisely the effect of breakup on fusion cross sections, it is necessary to be able to predict reliably the fusion cross sections expected in

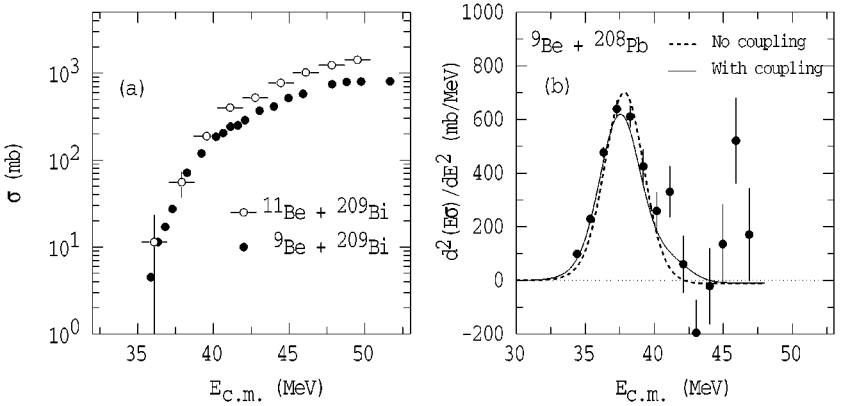


Figure 17 (*a*) Fusion cross sections for the reactions ^{9,11}Be + ^{209}Bi. Data are from Ref. 135. (*b*) The preliminary measured fusion barrier distribution for ^{9}Be + ^{208}Pb, together with scaled-down coupled-channels calculations, show that the average fusion barrier can be defined very precisely.

the absence of breakup. This is possible if the energy of the fusion barrier is unambiguously determined, and measurement of the fusion barrier distribution is clearly the best way to accomplish this. The distribution has recently been determined for the reaction ^{9}Be + ^{208}Pb (M Dasgupta, unpublished data) and is shown from preliminary data in Figure 17*b*. It consists of a single peak centered at $E_{c.m.} = 38$ MeV. The potential parameters can be determined from the peak position, which is stable against any reasonable energy dependence of the breakup probability, as the barrier distribution is narrow. The figure shows calculated barrier distributions with and without coupling to states in ^{208}Pb; as expected from the low charge product, the coupling is weak, and the shapes are very similar and are consistent with the data, suggesting that no unexpected coupling is present. However, the cross sections predicted in these calculations are considerably larger than those measured, indicating a substantial breakup probability of ^{9}Be. Products from incomplete fusion were also observed [(134); M Dasgupta, unpublished data] with large cross sections, showing that the ^{9}Be projectile breaks up mainly into two alpha particles and a neutron, with subsequent absorption of one or two of the fragments. Measurements of the breakup cross section where no fragments are absorbed by the target should produce a complete picture of breakup for this reaction.

Breakup of halo nuclei is expected to result primarily in the loss of the halo neutrons. Depending on the kinematics, and the radius at which breakup occurs, these can still be absorbed by the target nucleus. Thus the breakup may not

lead to a significant change in the fusion products, perhaps only resulting in a slight reduction of the average neutron number of the compound nucleus (135). Paradoxically, breakup of the stable ^{9}Be may be more significant than breakup of the unstable ^{10,11}Be in influencing the products of fusion. If confirmed, this conclusion is favorable for the fusion of unstable beams to form new, neutron-rich nuclei.

14. CONCLUSIONS AND OUTLOOK

Prior to the introduction of the concept of the experimental barrier distribution and the first measured distributions, fusion excitation functions were thought to be smooth and featureless curves, providing a poor test of theoretical models (136, 137). Since then, it has been demonstrated that in fact fusion excitation functions exhibit subtle structure that carries unique information on the interactions of the two nuclei at the fusion barrier. This is most easily visualized by taking the second derivative of the product $E\sigma$, which was shown within certain limits to represent the distribution of barrier probability with energy. Thus the fusion barrier distribution has been extended from an underlying concept believed to cause sub-barrier fusion cross section enhancement, to a quantity that can in principle be measured precisely. Interpretation of the new precise data has pushed forward the theoretical frontiers of the fusion process, as realistic calculations are required.

The barrier distribution representation removes a trivial geometrical effect from the experimental data. It reveals the quantity of greatest physical significance and interest and allows this quantity to be shown on a linear scale.

The weakness of the barrier distribution representation lies at the lowest beam energies, where transfer couplings may result in the lowest barrier having a very low weight. Here, the data and calculations are best represented in the traditional logarithmic plot of cross section against energy, from which the energy, if not the weight, of the lowest barrier is most clearly seen.

Although theoretically the concept of a barrier distribution is only valid under certain approximations, the experimental data shown here indicate that the situation in heavy-ion fusion is generally such that a barrier distribution is a meaningful concept, and that D_{exp} has physical significance.

Experimental distributions have shown that collective excitations generally play a dominant role in the fusion process, although coupling to positive *Q*-value transfer channels, if present, determines the energy of the lowest barrier. The barrier distributions currently existing can largely be described in terms of the collective properties of the separated nuclei. For systems with high $Z_1Z_2\beta$ that do not behave as good rotors, Coulomb excitation at large distances can

significantly affect the weights of the barriers (suppressing the lowest barriers) due to the opposite sign of the Coulomb and nuclear interaction. In a reaction involving different neutron excesses in projectile and target, and high charge product, multiple neutron transfers may result in a substantial reduction in energy of the lowest fusion barriers, which still carry considerable weight, thus dramatically increasing the sub-barrier fusion cross sections. This can be considered a collective mode, not of the separated nuclei, but of the combined system, namely neck formation.

Despite the advances in understanding fusion reactions, brought about partly by the new experimental evidence and partly by the theoretical impetus these data have provided, many questions remain open.

Fits to the precise cross section data at above-barrier energies generally require the diffuseness of the nuclear potential to be larger than that determined from elastic and inelastic scattering measurements. Heavy-ion fusion, however, probes the nuclear potential at smaller radii than does elastic scattering, so this may not be the contradiction it seems. Nevertheless, it remains to be determined whether such large a values reflect the nuclear potential or whether they come about through neglect in the calculations of significant physical effects, such as competetion with peripheral processes, couplings to high lying states, and effects of high angular momentum.

A convenient theory incorporating single and multinucleon transfer couplings is still elusive. It would be desirable to include transfer couplings in fusion models without requiring detailed spectroscopic information for each channel. Development in this direction is necessary to understand the transition from detailed coupled-channels calculations to macroscopic models of neck formation.

Further complete measurements of all other reaction channels need to be made for carefully chosen systems. Along with those systems for which such data already exist, they should be analyzed in a consistent (rather than fragmented) way, to investigate whether the current picture of the reaction dynamics is sufficient to describe such data. Although such a painstaking analysis may be time-consuming, improved computing speed makes it a more tractable project than it has been. Such complete analyses may even allow more reliable spectroscopic information to be obtained.

The precisely measured fusion cross sections are in principle a source of accurate information on spin distributions. Comparison of this information with that extracted from decay properties of the compound nucleus (gamma-ray multiplicities or fission anisotropies), and with measurement of (complex) peripheral reaction products, particularly for reactions with large charge product, should provide information on the competition between these processes as a

function of angular momentum. The interplay of fusion barrier distributions and compound nucleus spin distributions can be further exploited in the study of nuclear structure physics and of fission.

With the emergence of radioactive beam facilities, fusion with radioactive nuclei will give new opportunities to form exotic nuclei and to study new effects such as neutron skins, new collective modes (e.g. the pygmy resonance), and breakup effects.

For heavy systems with $Z_1Z_2 \geq 1600$, a comparison of the "capture" barrier distributions with those for true fusion (compound nucleus formation) should reveal more details of the extra-push effects. By a selective choice of appropriate systems, it may be possible to elucidate the contribution of Coulomb excitation in the entrance channel to the extra-push energies. The improved understanding of fusion may result in the development of novel routes to the elusive island of superheavy nuclei.

These open questions ensure continuing strong activity in the field of heavy-ion fusion, as both precise experiments and realistic theoretical calculations refine our understanding of the complex process involved.

ACKNOWLEDGMENTS

We thank our experimental and theoretical colleagues for their contributions to the work discussed in this review. In particular, we would like to highlight the role of Jack Leigh, who immediately recognized the significance of the experimental barrier distribution and made the first measurements so successful through his attention to detail and his thorough approach. The assistance of K Hagino, CR Morton, and H Timmers in preparing aspects of this review is gratefully acknowledged.

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